Endogenous Strength in Competition Between and Within Groups*

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Abstract

In this paper, I study competition between and within groups. I develop a model and design an experiment in which subjects play a two-stage contest: a group stage contest followed by an individual stage contest between winning group members. The experiment consists of different treatments in which a subject's strength in the individual stage may be linked positively or negatively with their contributions in the group stage. I find that the nature of this link affects how contributions are allocated over the two stages. I also observe how miscoordination in the group stage endogenously creates contests with asymmetric strengths in the individual stage. Ultimately, I find that subjects compete more aggressively in the individual stage when strength is negatively linked with group stage contributions versus when it is positively linked.

Keywords: group behavior, contests, experiment, personnel economics

JEL: C72, C91, C92, M54

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1 Introduction

Group work is ubiquitous in firms, sports, politics, and research. According to Cross et al. (2016), employees in some companies spend as much as 80% of their time on collaborative activities. Group mates work together to achieve a common goal or to compete against other groups for prizes. But competition also exists within groups. For example, employees in a firm work together to compete with rival firms for contracts and market share, and then compete internally for promotions and bonuses. Cyclists in breakaway groups coordinate to outpace the peloton and then sprint against each other for the finish line. After defeating the Axis powers in World War II, Allied nations clashed over spheres of influence in Europe. These examples all follow the same framework: an intergroup competition followed by an intragroup competition. Within this setting, group members choosing how much to contribute to their group anticipate the fact that they'll be competing with their fellow group members in the future. In many cases, the conflicts between the group and within the group are explicitly linked. An individual's strength or position in the intragroup competition often depends on how much they contributed to the intergroup competition. Helping to strengthen the group may make the individual either stronger or weaker when it comes time to compete with fellow group members.[1]

In this paper, I study competition between and within groups where an individual's strength in intragroup competition is linked to their contribution in the intergroup competition. I develop a model and conduct an experiment in which subjects play a two-stage contest,[2] and I systematically vary how contribution in the first stage affects strength in the second stage.

In my model, groups compete in two stages for an indivisible prize. The first stage (the group stage) is a standard group Tullock contest[3] (Katz et al., 1990; Münster, 2009). Each agent receives a fixed endowment and chooses how much to contribute to the group. Individual contributions are aggregated and the winning group is determined. Winning group members advance to the second stage (the individual stage) where they compete in a standard Tullock contest[4] (Tullock, 1967, 1980; Rogerson, 1982) to determine who ultimately wins the prize. In this stage, each agent's contributions are multiplied by their strength value, and the likelihood of winning is proportional to multiplied contributions. Strength is determined endogenously as a function of group stage contribution. I consider two different strength functions: one in which strength increases in group stage contribution and one in which it decreases. As benchmarks, I also consider two cases in which strength is exogenous to the group stage — one in which strengths are fixed and ex ante symmetric across all agents and another in which strength is randomly drawn and assigned for each agent from a known distribution. The model essentially combines two standard models, with one key novel innovation: the endogeneity of individual strength, which imposes a link between the two stages that can be systematically manipulated.

[1] For example, in a firm, the effort an employee devotes to attaining clients might affect her in the internal competition for promotions and bonuses. Depending on the organization, working hard to pursue a client might affect an employee's prospects of earning promotion positively or negatively. In one case, doing background research on clients and delivering pitches might make an employee a stronger candidate for internal promotion. In another case, the strongest employee may be the one who privately lobbies the firm's superiors, not necessarily the one who performs the productive tasks.

[2] Contests are frequently used to model competitive situations in firms, sports, lobbying, and R&D where output or likelihood of success is noisy and investments are unobservable or non-contractible (Tullock, 1967, 1980; Rogerson, 1982).

[3] See Sheremeta (2018) for a review of group contests.

[4] See Dechenaux et al. (2015) for a review of contests.

I characterize equilibrium contributions in the endogenous strength cases and the benchmark cases and develop hypotheses based on theoretical predictions.

I then conduct a controlled laboratory experiment in which two groups of two subjects each play the two-stage contest described above. In a between-subjects design across four treatments, I vary the relationship between group stage contribution and individual strength. In the **Increasing** treatment, higher group stage contributions increase strength. In the **Decreasing** treatment, higher group stage contributions decrease strength. I also run two benchmark treatments where strength is exogenous: **Symmetric** where each agent receives an equal strength ex ante, and **Random** where strength is drawn stochastically for each agent. Subjects participate in 20 repeated rounds with random rematching of groups between rounds. I collect data on contributions in each stage to compare against theoretical predictions and across treatments. I devote special attention to analyzing treatment differences in the individual stage. In my experimental design, I set up a unique environment in which the individual stage results are robust to risk attitudes, social preferences, bounded rationality, joy of winning, confusion, norms, and many other explanations. Due to the nature of the individual stage across all treatments, differences in contributions cannot be rationalized by these common explanations. With this in mind, I study how varying the link between the stages affects individual stage behavior.

A laboratory experiment allows us to empirically study this environment in ways that would be otherwise intractable using naturally occurring data. Real-world contest contributions are often undisclosed or otherwise difficult to observe. Natural settings also preclude control over parameters that drive interesting comparative static predictions. In this particular case, field settings that differ in the link between group contribution and individual strength also presumably differ in other significant ways. A lab setting allows us to manipulate the link between the two stages while holding all else constant and study the causal impacts on behavior.

I find two main sets of results. First, between treatments, aggregate contributions across both stages are the same, but contributions within each stage differ significantly. Within the group stage, contributions follow the pattern predicted by theory. Subjects contribute the most when contributions increase next-stage strength and the least when contributions decrease strength. Within the individual stage, contributions are significantly higher when strength is decreasing versus when it is increasing in group stage contributions. Based on aggregate effort in the two-stage contest, the cases I study appear to be equally efficient or profitable. However, due to the stark differences within each stage, a particular case may be considered optimal for a principal or social planner if either the collective or individual effort is considered productive. For example, a firm may want to incentivize collective effort but discourage unproductive infighting. On the other hand, patent races involve significant duplication of effort between research teams, with the most valuable innovation often emerging from internal competition. These results extend research on optimal contest and tournament design to a two-stage setting with endogenous strength (Tsoulouhas et al., 2007; Fu & Lu, 2012).

The second main set of results focuses on behavior in the individual stage. According to the theory, equilibrium contributions in the individual stage contest are fully determined by the ratio of realized strengths between contestants. Consequently, for a given strength ratio, the theory predicts equivalent contributions across all treatments. I find that differences in contributions are robust to controlling for strength ratio. Specifically, group members compete less aggressively with each other when strength stems from high contributions to the group and more aggressively when it stems from low contributions. Contrary to the theory, subjects care not only about the degree of strength asymmetry but also about the origins of that asymmetry.

The behavioral pattern I observe resembles previous findings on entitlement effects in dictator games (Hoffman et al., 1994; Cherry et al., 2002; Cappelen et al., 2007; Oxoby & Spraggon, 2008). An entitlement effect describes how individuals behave less prosocially when they earn an advantaged position. I extend this notion to a competitive setting and show how both the strong and weak players in a contest compete more aggressively when the strong position is earned through low contributions versus high contributions to the group. My results also expand our understanding of discouragement effects in asymmetric contests.[5] The discouragement effect describes a common tendency for contributions in a contest to fall when contestants are asymmetric. I find that the discouragement effect survives in my experiment only when higher strength comes from high contributions but disappears when higher strength comes from low contributions. In the latter case, both weak and strong players maintain relatively high levels of contributions, suggesting that discouragement effects are sensitive to how asymmetry arises.

This paper relates most directly to a small branch of the theoretical contest literature which studies competition between and within groups (Wärneryd, 1998; Katz & Tokatlidu, 1996). These models analyze the effects of varying group sizes and numbers of groups or the timing of competition (Münster, 2007). Extensions of the basic two-stage models study various types of asymmetry within or between groups (Choi et al., 2016; Stein & Rapoport, 2004). In my paper, I provide one of the few experimental tests of this setting, and I am the first to study an endogenous link between intergroup and intragroup competition.

Most closely related to my paper is one recent experiment that studies an intragroup contest followed by an intergroup contest (Bhattacharya & Rampal, 2024). In their experiment, group members first conflict over internal prize allocation before competing with other groups. This ordering of stages captures situations such as political elections where party members first compete in primaries for nominations, and then one candidate represents the party in the general election. In contrast to Bhattacharya & Rampal (2024), I study situations in which the intergroup competition precedes the intragroup competition. My setup focuses on situations in firms, sports, and research teams in which group members must first coordinate to defeat other groups before competing internally over rents. Also distinct from Bhattacharya & Rampal (2024), I focus on establishing an endogenous link between group and individual competition and explore how varying this link affects behavior.

The broader literature on multistage contests studies elimination tournaments (Gradstein & Konrad, 1999; Altmann et al., 2012; Sheremeta, 2010b) or situations in which investments in a contest carry over to future competition (Schmitt et al., 2004; Sheremeta, 2010a,b; Ryvkin, 2011). In a similar spirit, group stage contributions in my model affect an individual's position in the subsequent individual stage. My individual stage also borrows from previous contest models in which contestants possess asymmetric strength (or ability or effectiveness) (Stein, 2002; Baik, 1994; Schotter & Weigelt, 1992; Kirkegaard, 2012; Franke et al., 2013). While these models exogenously impose strengths, I study a unique setting in which strength is determined endogenously in different ways. I compare my findings to previous experimental work on asymmetric contests (Fonseca, 2009) and highlight how behavior differs when strengths are determined endogenously.

[5] See Dechenaux et al. (2015), Mealem & Nitzan (2016), and Chowdhury et al. (2023) for recent surveys.

2 Model

Consider m groups $g \in \{1, ..., m\}$ of n risk-neutral agents $i \in \{1, ..., n\}$ who compete in a two-stage contest for a known and common-value prize $V > 0$. Each agent ig receives an endowment $w > 0$ from which she may make contributions over both stages. In the first stage, the group stage, groups compete in a group contest. One group wins and advances to the second stage, the individual stage, in which members of the winning group compete for the prize. In the individual stage, each agent possesses strength θ_{ig} which may depend endogenously on group stage contributions.

2.1 Group Stage

In the group stage, m groups compete for the right to advance to the individual stage and compete for a prize. Each agent chooses how much of her endowment to contribute to the group contest. Individuals simultaneously select group stage contribution x_{ig}, and a group's contribution is the sum of its members' group stage contributions $X_g = \sum_{i=1}^{n} x_{ig}$. Any portion of an agent's endowment that she doesn't contribute in the group stage, $w - x_{ig}$, remains with her if she advances to the individual stage. Once all groups submit their contributions, one winning group is determined stochastically. According to a group Tullock lottery contest success function (Münster, 2009; Skaperdas, 1996), group g wins with the following probability:

$$p_g = \begin{cases} \frac{X_g}{\sum_{h=1}^{m} X_h} & \text{if} \quad \sum_{h=1}^{m} X_h > 0 \\ \frac{1}{m} & \text{otherwise} \end{cases} \tag{1}$$

Agents earn no reward in the group stage. Members of the winning group advance to the individual stage where they have the opportunity to compete against one another for the prize V.

2.2 Individual Stage

In the individual stage, each member of the winning group chooses how much of her remaining endowment to contribute to the individual contest. Individuals simultaneously select individual stage contributions y_{ig}. Individual stage contributions are then multiplied by strength. Multiplied contributions are used to determine the winner in the individual stage. Given multiplied contributions of both agents, agent ig wins with the probability captured by the following contest success function (Baik, 1994; Stein, 2002; Skaperdas, 1996):

$$p_{ig} = \begin{cases} \frac{\theta_{ig} y_{ig}}{\sum_{j=1}^{n} \theta_{jg} y_{jg}} & \text{if} \quad \sum_{j=1}^{n} \theta_{jg} y_{jg} > 0 \\ \frac{\theta_{ig}}{\sum_{j=1}^{n} \theta_{jg}} & \text{otherwise} \end{cases} \tag{2}$$

2.3 Payoffs

Each agent earns her endowment minus her group stage contribution. Members of the winning group also subtract their individual stage contributions. Ultimately only one agent out of four earns the prize V. This agent belongs to the winning group and wins the individual stage. Given the above probabilities of winning in each stage, the expected payoff for the two-stage contest for agent ig is:

$$\pi_{ig} = w - x_{ig} + p_g(p_{ig}V - y_{ig}) \tag{3}$$

2.4 Strength

The main innovation of the model is the freedom to vary the endogeneity and asymmetry of strengths in the individual stage. Through the strength parameter, I establish a novel link between the group stage and the individual stage. In the individual stage, an agent's strength is a function of her group stage contribution. In general, $\theta_{ig} = f(x_{ig})$ where f represents the *strength function*. I consider two cases in which strength is endogenous to group stage contribution. In one case f is monotonically increasing and concave in x_{ig}. Higher group stage contributions translate into higher strength. In another case, f is monotonically decreasing and concave in x_{ig}. In this case, the more an agent contributes to her group, the lower her strength in the individual stage.

As a baseline, I also consider when strength is exogenous and symmetric across all subjects such that $\theta_{ig} = c \quad \forall \quad ig$. In this case, regardless of group stage decisions, strength is known, constant and equivalent across all agents. Finally, I consider an additional benchmark case in which strength is exogenous and asymmetric. For each subject, a strength is drawn randomly from a uniform distribution on the interval [1,5]. In the analysis that follows, I discuss equilibrium and comparative statics for these cases.

2.5 Analysis

I now characterize subgame-perfect equilibria of the two-stage contest with potentially endogenous strength. In order to make the analysis more tractable and derive closed-form predictions for an experiment, I focus on the version of the model with $m = 2$ groups and $n = 2$ agents. The model is solved via backward induction. Without loss of generality, fix the winning group in the individual stage to be group a and define

$$\theta_{ia} = f(x_{ia}), \qquad r_{ia} = w - x_{ia}.$$

where r_{ia} denotes the remaining endowment. Each agent $ia \in \{1a, 2a\}$ chooses $y_{ia} \in [0, r_{ia}]$ to maximize

$$u_{ia}(y_{ia}, y_{ja}) = \frac{\theta_{ia}y_{ia}}{\theta_{1a}y_{1a} + \theta_{2a}y_{2a}}V - y_{ia} + r_{ia}.$$

Lemma 1 (Individual stage Nash equilibrium)**.** *Assuming non-binding budget constraints,*[6] *given any vector of strengths* $(\theta_{1a}, \theta_{2a})$*, the individual stage has a unique symmetric Nash equilibrium in which*

[6]For a discussion of the individual stage Nash equilibrium when budget constraints bind, see Appendix A.1.

$$y_{1a}^* = y_{2a}^* = \frac{V\theta_{1a}\theta_{2a}}{(\theta_{1a}+\theta_{2a})^2}.$$

Winning probabilities are

$$p_{1a}^* = \frac{\theta_{1a}}{\theta_{1a}+\theta_{2a}}, \qquad p_{2a}^* = \frac{\theta_{2a}}{\theta_{1a}+\theta_{2a}},$$

and continuation payoffs are

$$U_{1a} = \frac{V\theta_{1a}^2}{(\theta_{1a}+\theta_{2a})^2}, \qquad U_{2a} = \frac{V\theta_{2a}^2}{(\theta_{1a}+\theta_{2a})^2}.$$

Proof. See Appendix A.2 □

Notice that the individual stage Nash equilibrium is symmetric even if agents are asymmetric, but equilibrium effective contributions (contributions multiplied by strength), probabilities of winning, and expected payoffs differ between the strong and weak agent.

Figure 1 illustrates how the individual stage Nash equilibrium changes with strengths using example parameters that match the parameters I use in the experiment. The horizontal axis is the strength ratio, which equals the lower strength in the group divided by the higher strength in the group. A strength ratio of 1 means perfect symmetry between agents. Notice that the stage Nash equilibrium is maximized whenever agents are symmetric and decreases as agents become more asymmetric. The intuition is that higher inequality between agents dampens competition and puts downward pressure on contributions. The strong agent can achieve the same probability of winning with a lower contribution, so he finds it optimal to decrease contributions relative to the equilibrium when both agents are equally strong. The weak agent finds it optimal to decrease contributions as well. The greater the wedge between the strong and weak agent, the lower the equilibrium contribution level.

Importantly, strengths should be taken as given by both agents in the individual stage, and the equilibrium remains identical regardless of how strengths were determined. Therefore, whether strengths are randomly assigned or endogenously earned by a prior decision should not affect the individual stage equilibrium as long as the budget constraint doesn't bind.

Remark 1 (Strength assignment)**.** *The individual stage Nash equilibrium depends only on the vector of strengths and not on how strengths are assigned.*

Moving backwards to the group stage and substituting individual stage continuation values conditional on winning, the expected payoff of agent ig in the group stage is

$$\pi_{ig}(x_{ig}, x_{jg}, X_{-g}) = w - x_{ig} + p_g(X_g, X_{-g})\, U_{ig}\big(f(x_{ig}), f(x_{jg})\big) \tag{4}$$

Agent ig chooses $x_{ig} \in [0, w]$ to maximize expected two-stage payoff. I first consider symmetric equilibrium group stage actions.

Lemma 2 (Symmetric interior equilibrium condition)**.** *Any symmetric interior group stage equilibrium $x^* > 0$ satisfies*

$$1 = \frac{V}{32x^*} + \frac{V}{8}\frac{f'(x^*)}{f(x^*)}.$$

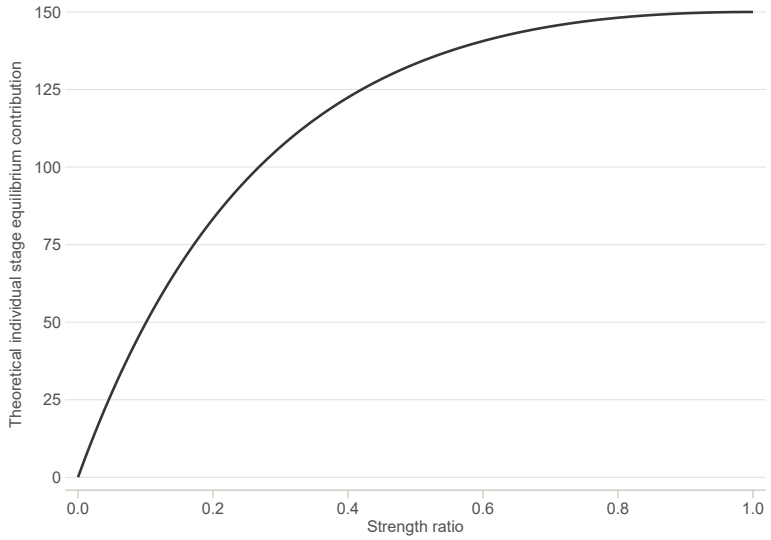


Figure 1: Theoretical Individual Stage Equilibria by Strength Ratio, Experimental Parameters

Notes: Theoretical individual stage Nash equilibria for given strength ratios using experimental parameters ($w = 300$, $V = 600$). Strength ratio equals the lower strength in the group divided by the higher strength in the group.

Proof. See Appendix A.2 □

This condition highlights the two forces governing group stage investment. The first term reflects the effect of contributions on the probability of advancing to the individual stage, while the second term captures the effect of contributions on strength and thus the continuation value in the individual stage.

2.5.1 Exogenous Symmetric Strength

As a benchmark, suppose strength is exogenous and symmetric across all agents:

$$\theta_{ig} = \theta > 0 \qquad \forall\, i, g.$$

Then the individual stage (conditional on reaching it) is a standard symmetric two-player Tullock contest. By Lemma 1, the unique (interior) Nash equilibrium of the individual stage satisfies

$$y^*_{1g} = y^*_{2g} = \frac{V}{4}, \qquad p^*_{1g} = p^*_{2g} = \frac{1}{2}, \qquad u_{1g} = u_{2g} = \frac{V}{4},$$

where $u_{ig} = p^*_{ig} V - y^*_{ig}$ denotes the continuation payoff conditional on the group winning the group stage (net of the constant $r_{ig} = w - x_{ig}$ term).

Substituting $u_{ig} = V/4$ into the group stage payoff, agent ig's expected two-stage payoff in the group stage becomes

$$\pi_{ig}(x_{ig}, x_{jg}, X_{-g}) = p_g(X_g, X_{-g})\frac{V}{4} - x_{ig} + w, \tag{5}$$

where X_{-g} is the other group's total contributions.

Proposition 1 (Multiple subgame-perfect equilibria with exogenous strength)**.** *Assume $\theta_{ig} = \theta > 0$ for all ig and that individual stage budget constraints do not bind. Then there is a continuum of subgame-perfect equilibria. In any such equilibrium,*

$$X_a^* = X_b^* = \frac{V}{16},$$

and within each group $g \in \{a, b\}$ any split of the total group contribution is consistent with equilibrium:

$$x_{1g}^* \in \left[0, \frac{V}{16}\right], \qquad x_{2g}^* = \frac{V}{16} - x_{1g}^*.$$

In the individual stage, the unique equilibrium is $y_{1g}^* = y_{2g}^* = V/4$.

Proof. See Appendix A.2 □

2.5.2 Endogenous Strength

In the general model, the strength function f is unrestricted. However, without additional structure, the reduced group stage game may admit multiple equilibria, including asymmetric equilibria, which complicates both theoretical characterization and empirical predictions. To obtain sharp and tractable predictions, I restrict attention to the following class of strength functions:

$$f(x) = (x + \alpha)^\gamma, \qquad \alpha > 0,\ \gamma \in (-1, 1) \setminus \{0\}.$$

This class of functions is strictly positive and continuously differentiable on $[0, w]$, ensuring well-defined contest success functions and facilitating backward induction. When $\gamma \in (0, 1)$, strength is increasing and exhibits diminishing returns, while when $\gamma \in (-1, 0)$, strength is decreasing in group stage contributions.

These strength functions permit tractable equilibrium characterization, allowing for closed-form solutions and transparent comparative statics. The experimental treatments correspond to two particular cases within this class: $f(x) = (x + 3)^{1/3}$, which features positive monotonically increasing concave strength, and $f(x) = (x + 3)^{-1/3}$, which features positive monotonically decreasing convex strength. These cases generate distinct incentives linking the group stage to the individual stage while remaining analytically tractable. The bounded curvature implied by $\gamma \in (-1, 1)$ ensures well-behaved equilibrium behavior and, in the cases studied in the experiment, yields a unique equilibrium.

Proposition 2 (Unique symmetric subgame-perfect equilibrium with endogenous strength)**.** *Assume individual stage budget constraints do not bind and let*

$$f(x) = (x + \alpha)^\gamma, \qquad \alpha > 0,\ \gamma \in (-1, 1) \setminus \{0\}.$$

Then the two-stage game admits a unique symmetric subgame-perfect equilibrium. In this equilibrium, group stage contributions are symmetric and equal across all agents,

$$x_{1a}^* = x_{2a}^* = x_{1b}^* = x_{2b}^* = x^* > 0,$$

The symmetric group stage contribution x^* is the unique solution in $(0, w]$ to

$$1 = \frac{V}{32x} + \frac{V}{8}\frac{f'(x)}{f(x)} = \frac{V}{32x} + \frac{V\gamma}{8(x + \alpha)}, \tag{6}$$

and is given in closed form by

$$x^* = \frac{V(1+4\gamma) - 32\alpha + \sqrt{(32\alpha - V(1+4\gamma))^2 + 128V\alpha}}{64}. \tag{7}$$

Individual stage contributions are given by the unique Nash equilibrium characterized in Lemma 1 for any vector of strengths.

Proof. See Appendix A.2 □

Corollary 1 (No asymmetric group stage equilibrium in special cases)**.** *Assume individual stage budget constraints do not bind and let*

$$f(x) = (x+3)^{\gamma}, \qquad \gamma \in \left\{\tfrac{1}{3}, -\tfrac{1}{3}\right\}.$$

Then a subgame-perfect equilibrium does not admit asymmetric group stage contributions. In particular, in any equilibrium,

$$x^*_{1a} = x^*_{2a} = x^*_{1b} = x^*_{2b} = x^* > 0,$$

Proof. See Appendix A.2 □

Corollary 2 (Comparative statics across experimental treatments)**.** *Assume budget constraints do not bind. Consider the following treatments:*

1. ***Increasing strength (INC):*** $f(x) = (x+3)^{1/3}$.
2. ***Decreasing strength (DEC):*** $f(x) = (x+3)^{-1/3}$.
3. ***Symmetric strength (SYM):*** $f(x) = \theta > 0$.

Let x^*_T denote the symmetric equilibrium group stage contribution of each agent under treatment $T \in \{INC, SYM, DEC\}$, and let y^*_T denote the symmetric equilibrium individual stage contribution of each agent. Then:

1. ***Group stage contributions.***

$$x^*_{INC} > x^*_{SYM} > x^*_{DEC}.$$

2. ***Individual stage contributions.***

$$y^*_{INC} = y^*_{SYM} = y^*_{DEC}.$$

3. ***Total expenditure.*** *Fix a as the winning group and let total expenditure be defined as*

$$TOT^*_T = \frac{X_a + y_{1a} + y_{2a}}{V}.$$

Then

$$TOT^*_{INC} > TOT^*_{SYM} > TOT^*_{DEC}.$$

Proof. Follows directly from Proposition 1 and Proposition 2. □

2.5.3 Exogenous Asymmetric Strength

Finally, I consider the case in which strength is exogenous but asymmetric. Suppose each agent's strength θ_{ig} is independently drawn from a continuous uniform distribution on $[1, 5]$. In the group stage, each agent observes only her own strength and knows the distribution from which all strengths are drawn. An agent does not observe the strength of her group mate. Once a group advances to the individual stage, both members observe realized strengths. Lemma 1 characterizes the unique equilibrium in the individual stage and the corresponding continuation payoffs. Conditional on group g winning the group stage, agent ig's continuation payoff is

$$U_{ig} = \frac{V\theta_{ig}^2}{(\theta_{ig} + \theta_{jg})^2}. \tag{8}$$

Differentiating with respect to own strength yields

$$\frac{\partial U_{ig}}{\partial \theta_{ig}} = \frac{2V\theta_{ig}\theta_{jg}}{(\theta_{ig} + \theta_{jg})^3} > 0, \tag{9}$$

which indicates that continuation payoffs in the individual stage increase in own strength.

Moving backwards to the group stage, an agent observes her own strength but not the strength of any other player. Let $x(\theta_{ig})$ denote a Bayesian equilibrium contribution strategy, where $x(\theta_{ig})$ specifies the group stage contribution of an agent with strength θ_{ig}. Consider agent $1a$ who observes θ_{1a} and chooses $x(\theta_{1a}) \in [0, w]$ to maximize

$$\pi_{1a} = w - x(\theta_{1a}) + \mathbb{E}\left[\frac{x(\theta_{1a}) + x(\theta_{2a})}{x(\theta_{1a}) + x(\theta_{2a}) + x(\theta_{1b}) + x(\theta_{2b})} \frac{\theta_{1a}^2}{(\theta_{1a} + \theta_{2a})^2}\right], \tag{10}$$

where the expectation is taken over all possible draws of the unobserved strengths. Differentiating expected payoff with respect to $x(\theta_{1a})$ yields

$$1 = \mathbb{E}\left[\frac{x(\theta_{1b}) + x(\theta_{2b})}{(x(\theta_{1a}) + x(\theta_{2a}) + x(\theta_{1b}) + x(\theta_{2b}))^2} \frac{\theta_{1a}^2}{(\theta_{1a} + \theta_{2a})^2}\right]. \tag{11}$$

where the right hand side represents the marginal benefit of increasing group stage contribution.

I do not offer a closed-form characterization of the equilibrium contribution function $x(\theta_{ig})$ which specifies the optimal group stage contribution for every strength. Instead, I make one remark about the relationship between observed own strength and equilibrium group stage contribution to guide predictions about behavior in the experiment. Since continuation value increases in own strength, agents who draw a higher strength face a higher expected reward from winning the group stage and advancing to the individual stage. Consequently, equilibrium group stage contribution increases in observed own strength.

Remark 2 (Group stage contributions and strength)**.** *When strength is randomly drawn for each agent from a uniform distribution, group stage contributions are strictly increasing in strength. That is, for* $\theta', \theta \sim U[1, 5]$ *with* $\theta' > \theta$*,*

$$x(\theta') > x(\theta). \tag{12}$$

Proof. See Appendix A.2 □

Essentially, higher strength translates to higher individual stage continuation value and higher marginal benefit of increasing group stage contribution. Therefore, group stage contribution increases in strength.

3 Experimental Design

I design and conduct a laboratory experiment in which subjects play the two-stage contest outlined above. I employ a between-subjects design across four treatments where I vary the relationship between group stage contribution and individual strength. In the **Increasing** treatment, strength increases in group stage contribution. In the **Decreasing** treatment, strength decreases in group stage contribution. In the baseline **Symmetric** treatment, strength is exogenously given and symmetric. Finally, I run an additional **Random** treatment in which strength is exogenously given and randomly drawn for each subject from a uniform distribution on the interval [1,5].[7] Table 1 summarizes the experimental sessions, treatments and total participants. I run a total of 12 sessions consisting of 16 subjects per session for a total of 192 subjects (55% female). All experimental sessions take place at the XS/FS Lab at Florida State University and last at most 90 minutes. Each subject participates in exactly one session with no prior experience. I implement the experiment using z-Tree (Fischbacher, 2007) and recruit subjects in ORSEE (Greiner, 2015). Subjects earn $19.86 on average including a $7 show-up fee.

Table 1: Summary of Experimental Treatments

Treatment	**Sessions**	**Participants**
Increasing	3	48
Decreasing	3	48
Symmetric	3	48
Random	3	48
	12	192

In each session, subjects first watch a video recording of the experimenter reading over the experiment instructions.[8] Each subject also receives a paper copy of the instructions. Before proceeding to the decision rounds, subjects complete a comprehension check quiz and participate in an unpaid 3-minute practice module that gives subjects practice with the experiment interface. Next, subjects complete the main incentivized part of the experiment involving a sequence of decision rounds. Finally, subjects complete a questionnaire consisting of unincentivized and optional questions. I elicit gender, risk preferences, and open responses on how subjects made their decisions.

In the main part of the experiment, subjects play a group contest followed by an individual contest. I create match groups of 8 subjects each and explain that subjects will only interact with members of their match group. Subjects play 20 repeated rounds of the two-stage contest. I randomly re-pair group mates and rematch groups within match groups between rounds. Each

[7] This treatment was not in the original preregistration. It was introduced after observing inititial data from the first three treatments.

[8] A copy of the instructions for **Increasing** can be found in Appendix C. Instructions for the other three treatments are available upon request.

decision round consists of two stages. At the beginning of each round, each participant receives a one-time endowment of 300 points which they may contribute over both stages in the round.

The first stage is the group stage. In the group stage, two groups of two subjects each participate in a group contest. Groups compete only for the right to earn a prize in the next stage. Each individual chooses a group stage contribution from their endowment, and a group's contribution is the sum of the group stage contributions of both members. Whatever a subject decides not to contribute in this stage remains with her for the next stage.

Subjects in each treatment understand whether and how group stage contributions affect strength. In **Increasing** and **Decreasing**, subjects are given a calculator to determine the strength that will result from a group stage contribution before making a final submission. Once a group stage contribution is submitted, subjects immediately see their own strengths. In **Random**, subjects understand that each subject's strength is drawn from a uniform distribution on the interval [1,5]. Each subject sees only her own strength on screen at the beginning of the group stage. However, subjects do not observe group mate strengths until the individual stage.[9]

One group wins the contest according to the group contest-success function described in the model above. However, all subjects advance to the next stage without learning the outcome of the group stage. Subjects see their contributions and their group mate's contributions as well as the corresponding remaining endowments. At this time, in **Increasing**, **Decreasing**, and **Random** each subject also learns the strength of her group mate.

The second stage is the individual stage. In the individual stage, subjects participate in the individual contest described in the model. The two group mates from the previous stage compete for a prize of 600 points. Subjects at this point don't know the outcome of the group stage and compete as if their group won. Each subject selects an individual stage contribution from her remaining endowment. Individual stage contributions are multiplied by strength, and multiplied contributions determine each subject's probability of winning the individual stage.

After subjects submit individual stage contributions, they see their group mate's decision and corresponding multiplied contribution and then learn the outcome of the round. At this point, I reveal the results of the group stage and the individual stage. Subjects who lose in the group stage earn their endowment minus their group stage contribution only. Subjects who win in the group stage but lose in the individual stage earn their endowment minus their group stage contribution and individual stage contribution. Finally, subjects who win in both stages earn an additional 600 points. After the last decision round, I randomly select 4 out of 20 rounds for payment. At the end of the experiment, I convert earnings in points to USD for payment at a rate of 1 point equals \$0.01.

I list testable hypotheses below which follow directly from the theoretical analysis in the previous sections. Since multiple equilibria exist in **Symmetric**, I focus on the symmetric equilibrium for predictions and comparisons. The prediction for group stage contributions in this treatment can also be interpreted as the average equilibrium contributions.

Hypothesis 1 (Group Stage)**.** *Group stage contributions are highest in* ***Increasing****, next highest in* ***Symmetric*** *and lowest in* ***Decreasing****.* $x^*_{INC} > x^*_{SYM} > x^*_{DEC}$*. In* ***Random****, group stage contributions are increasing in own strength.*

[9] Revealing only own strength most closely aligns with the information structure of the endogenous strength treatments in which subjects effectively choose own strength at the time of making a group stage contribution but don't yet know group mate's choice and corresponding strength.

Table 2: Summary Statistics, Last 10 Rounds

	Group Stage		Individual Stage	
Treatment	Mean	Prediction	Mean	Prediction
Increasing	60.35	42.09	121.03	150.00
	[39.24, 86.10]		[101.99, 138.24]	
Decreasing	17.48	4.19	160.73	150.00
	[3.75, 37.94]		[123.16, 194.17]	
Symmetric	32.82	18.75	174.10	150.00
	[22.79, 41.90]		[152.12, 188.44]	
Random	52.04	27.55	125.86	143.25
	[35.67, 67.14]		[109.04, 152.97]	

Notes: Mean contributions by stage and treatment with theoretical predictions. Data from last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.

Hypothesis 2 (Individual Stage)**.** *Individual stage contributions are equivalent across all treatments.* $y^*_{INC} = y^*_{SYM} = y^*_{DEC}$.

Hypothesis 3 (Total Expenditure)**.** *Total expenditure is highest in* ***Increasing****, next highest in* ***Symmetric*** *and lowest in* ***Decreasing****.* $TOT^*_{INC} > TOT^*_{SYM} > TOT^*_{DEC}$.

It is possible for subjects in different treatments to face similar vectors of strength in the individual stage. In theory, subjects should only care about the strengths themselves and not how those strengths were determined. Whether strengths were exogenously assigned or endogenously determined in a particular way shouldn't affect decisions. Strengths should be taken as given once the individual stage arises. Therefore, I propose an additional null hypothesis that individual stage behavior won't vary across treatments for a given vector of strengths. In the following section, I test this hypothesis and consider behavioral explanations to explain when differences arise across treatments contradicting the theoretical prediction.

Hypothesis 4 (Strength Assignment)**.** *For a given vector of strengths, individual stage contributions should not vary across treatments.*[10]

4 Results

I analyze aggregate results on group stage contributions and individual stage contributions from 192 subjects over 12 sessions and four treatments. Since I randomly rematch within match-groups each round, I conduct non-parametric tests on match-group averages as independent observations. I also employ linear regression analysis. Due to the small number of independent clusters, I report 95% wild cluster bootstrap confidence intervals on regression coefficients, and I conduct all statistical inference using p-values computed with the wild cluster bootstrap method (Cameron et al., 2008). I focus my main analysis on the last 10 rounds of the data.[11]

[10] This hypothesis was not included in the original preregistration.

[11] Results using all data are available upon request.

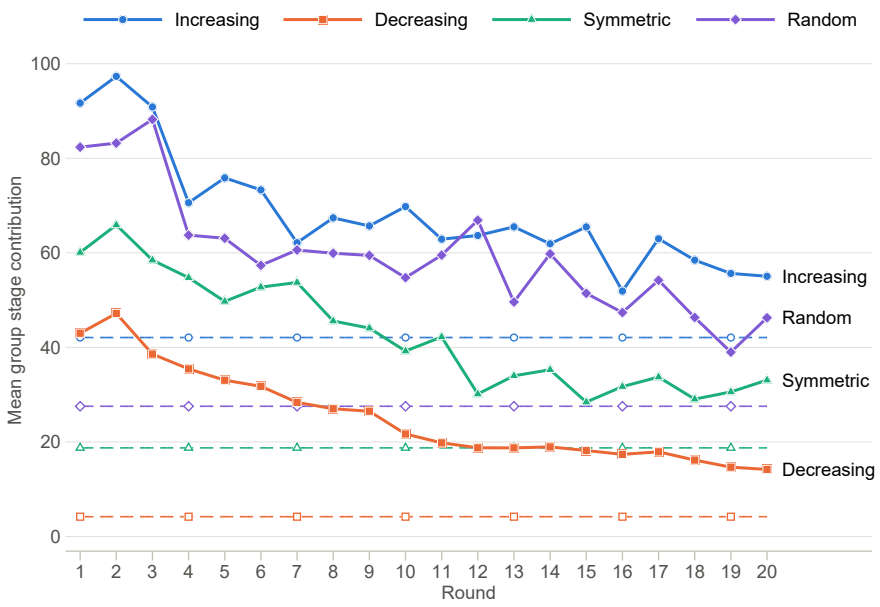


Figure 2: Group Stage Contributions by Round

Notes: Mean group stage contributions in each round by treatment. Dashed lines with markers show theoretical predictions for each corresponding treatment.

Table 2 reports summary statistics and theoretical predictions for contributions in each stage and each treatment over the last 10 rounds. Predictions in **Increasing**, **Decreasing**, and **Symmetric** are derived from subgame perfect equilibria. For **Symmetric**, multiple equilibria exist, but I focus on the symmetric equilibrium. The prediction for **Random** is a benchmark derived from the mean strength draws given the distribution of strength used in the experiment. Data from **Random** will primarily be used to study individual stage contributions in a later section.

4.1 Group Stage Contributions

Figure 2 depicts mean contributions by round in the group stage. It also includes the theoretical predictions for each treatment. Substantial adjustments take place over the first 10 rounds after which behavior stabilizes considerably. In figure 7 in Appendix B, I also plot the empirical CDF of contributions in the group stage for each treatment for the last 10 rounds.

In the group stage, contributions exceed the equilibrium predictions on average for each treatment.[12] However, overinvestment diminishes over the course of 20 repeated rounds (see figure 2).

Variations in group stage contributions across treatments follow the pattern predicted by the theory and in Hypothesis 1. According to pairwise Wilcoxon rank-sum tests on match-group averages, contributions are significantly higher in **Increasing** compared to each of the other

[12] Wald tests comparing constants in linear regression to predictions: INC, $p = 0.11$; DEC, $p = 0.06$; SYM, $p = 0.01$

treatments except **Random**. Contributions in **Symmetric** exceed contributions in **Decreasing**.[13] It appears subjects in each treatment anticipate the effect of group stage contributions on individual strength. As predicted by theory, group stage contributions in **Random** increase in own strength ($p = 0.03$; see table 6 in Appendix B).

Result 1 (Group stage contributions)**.** *Relative to subgame-perfect equilibrium predictions, subjects in all treatments overinvest in the group stage. Consistent with the theory and Hypotheses 1, group stage contributions are highest when the strength function is increasing and lowest when it is decreasing. Group stage contributions in* ***Random*** *increase in strength.*

4.2 Individual Stage Contributions

Figure 3 depicts mean contributions by round in the individual stage. The figure also includes the theoretical predictions for each treatment. Substantial adjustments take place over the first 10 rounds after which behavior stabilizes considerably. In figure 8 in Appendix B, I also plot the empirical CDF of contributions in the individual stage for each treatment for the last 10 rounds.

In the individual stage, the subgame-perfect equilibrium in each treatment predicts a contribution of 150 points. Contrary to the typical result found in individual Tullock contest experiments, I observe little overinvestment in this stage relative to the subgame-perfect equilibrium benchmark. Focusing on the last 10 rounds, participants only significantly overinvest on average in **Symmetric**. Contributions in **Increasing** actually fall significantly below equilibrium prediction on average.[14]

Of course, a better benchmark for testing overinvestment in the individual stage is the realized Nash equilibrium for each group. Along the equilibrium path, group stage contributions are symmetric. Off equilibrium path group stage contributions may induce asymmetry in the individual stage which dampens competition and puts downward pressure on the unique symmetric individual stage Nash equilibrium (see figure 1).

I calculate the realized individual stage Nash equilibrium for each group of two subjects given their group stage contributions and corresponding strengths.[15] I calculate percent deviations from the stage Nash equilibrium. In figure 9 in Appendix B, I plot mean percent deviations from stage Nash equilibrium by round for each treatment.[16] I find that the mean percent deviation in **Symmetric** is significantly greater than 0. The mean individual stage contribution in **Symmetric** over the last 10 rounds is 16.27% above the stage Nash equilibrium. The mean percent deviation from NE is actually negative for **Increasing** and **Random**. The mean individual stage contribution in **Increasing** over the last 10 rounds is 11.99% below the stage Nash

[13] Wilcoxon rank-sum tests: INC vs. DEC, $p = 0.009$; INC vs. SYM, $p = 0.04$; INC vs. $RAND$, $p = 0.59$; DEC vs. SYM, $p = 0.09$; DEC vs. $RAND$, $p = 0.009$; SYM vs. $RAND$, $p = 0.04$

[14] Wald tests comparing constants in linear regression to predictions: INC, $p = 0.01$; DEC, $p = 0.44$; SYM, $p = 0.04$

[15] Figure 10 plots the raw data on the realized strength ratio and corresponding stage Nash equilibrium for each group.

[16] I exclude contests in which at least one subject contributes the entire 300 point endowment in the group stage. In this case, the contestant who contributed everything must choose zero in the individual stage. The optimal best response for the other contestant is to contribute exactly one point. If this contestant submits something substantially larger than one, then the percent deviation is extreme and greatly affects the mean percent deviation calculations. Therefore, for this analysis, I drop cases in which subjects contribute everything in the group stage which only occurs twice in the last 10 rounds.

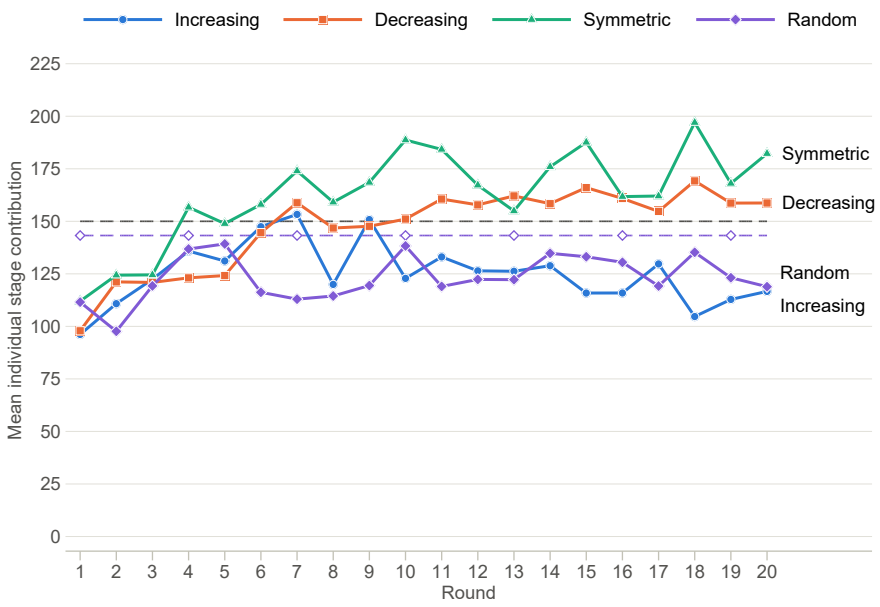


Figure 3: Individual Stage Contributions by Round

Notes: Mean individual stage contributions in each round by treatment. Flat black dashed line shows subgame perfect equilibrium prediction for Increasing, Decreasing, and Symmetric. Dashed purple line with diamonds shows theoretical prediction for Random.

equilibrium. This provides some suggestive evidence that when strength is increasing in group stage contributions, individual stage contributions fall not only below the subgame-perfect equilibrium prediction but also below the stage Nash equilibrium prediction. However, the difference from 0 percent deviation is not significant.[17] I also find that percent deviations are significantly lower in **Increasing** compared to **Decreasing** and **Symmetric** but no different compared to **Random**.[18] Many laboratory experiments involve subjects playing one-shot or repeated contests in isolation. Perhaps framing contests in dynamic environments that are more complex and realistic actually deters overinvestment.

According to the theory, despite differences in group stage contribution levels, individual stage contributions should be the same across treatments. Contrary to the theoretical predictions and Hypothesis 2, I find significant differences on average across treatments which emerge and persist in the last 10 rounds. I find mean contributions in **Increasing** significantly below **Decreasing** and **Symmetric**. I find no significant difference between **Decreasing** and **Symmetric** or between **Increasing** and **Random**.[19]

Result 2 (Individual stage contributions)**.** *In the individual stage, relative to the subgame perfect equilibrium or the realized Nash equilibrium, subjects do not overinvest on average in*

[17] Wald tests comparing constants in linear regressions to predictions: INC, $p = 0.14$; DEC, $p = 0.25$; SYM, $p = 0.04$; $RAND$, $p = 0.24$

[18] Wilcoxon rank-sum tests: INC vs. DEC, $p = 0.09$; INC vs. SYM, $p = 0.02$; INC vs. $RAND$, $p = 0.59$; DEC vs. SYM, $p = 0.94$; DEC vs. $RAND$, $p = 0.06$; SYM vs. $RAND$, $p = 0.009$

[19] Wilcoxon rank-sum tests: INC vs. DEC, $p = 0.06$; INC vs. SYM, $p = 0.02$; INC vs. $RAND$, $p = 0.70$; DEC vs. SYM, $p = 0.70$; DEC vs. $RAND$, $p = 0.06$; SYM vs. $RAND$, $p = 0.009$

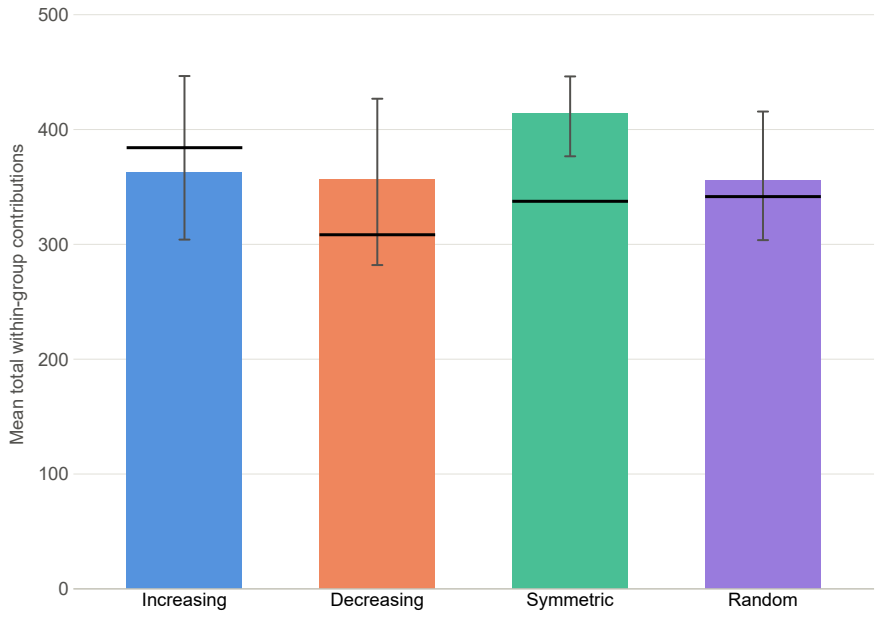


Figure 4: Within-Group Total Contributions, Last 10 Rounds

Notes: Mean total within-group contributions over both stages by treatment. Black lines show theoretical predictions. Data from last 10 rounds. Error bars represent 95% wild cluster bootstrap confidence intervals.

the endogenous strength treatments. Contrary to the theory and Hypothesis 2, individual stage contributions are lower when the strength function is increasing versus when it is decreasing.

The first two results analyze contributions in each stage in isolation. I now consider aggregate contributions over both stages before returning to the individual stage for additional analysis.

4.3 Total Contributions over Both Stages

So far, I observe significant differences between contributions in each stage in the endogenous strength treatments. Next, I consider aggregate contributions over both stages. I focus on total expenditures over both stages within a group. This statistic serves as a measure of market efficiency or rent dissipation.[20] Figure 4 plots mean within-group total expenditures for each treatment. The figure also includes equilibrium predictions and 95% wild cluster bootstrap confidence intervals around estimates.[21] Theoretically, any differences across treatments in within group total expenditures should be solely driven by differences in group stage contribution as a consequence of varying the strength function. Empirically, endogenous strength affects contributions in both stages and the effects in each stage go in opposite directions. Ultimately, I observe no significant differences between the endogenous strength treatments in either measure of efficiency.

[20] In figure 11 in Appendix B, I also compare expected payoffs across treatments.
[21] Tables showing mean estimates and confidence intervals are located in Appendix B.

Regarding within-group total expenditures, pairwise Wilcoxon rank-sum tests using the last 10 rounds of the data show no significant differences between treatments except for when comparing **Symmetric** to **Random**.[22] In particular, contrary to the theory and Hypothesis 3, no significant difference emerges in expenditures between **Increasing** and **Decreasing**. Compared to the equilibrium predictions, within group total contributions are higher than expected in **Symmetric** and **Decreasing** (significantly so in **Symmetric**) but lower than expected in **Increasing** on average.[23]

Result 3 (Total expenditures over both stages)**.** *Contrary to theory and Hypothesis 3, total expenditures over both stages do not differ significantly between treatments with endogenous strength. Varying the strength function does not affect total contributions, but it does affect the stage in which contributions are expended.*

In conjunction with Result 1 and Result 2, Result 3 is the first key contribution of the paper which offers potential implications for organizational and industrial policy. In the end, whether group stage contributions positively or negatively affect an individual's position in future internal conflict, total effort expenditure remains the same on average. However, the relationship between group effort and individual strength does substantially affect the timing and nature of effort expenditure. Specifically, when strength is increasing compared to when it is decreasing, group effort is higher and individual effort is lower.

If a manager or social planner cares only about aggregate effort expenditure, the incentive structure relating group competition and individual conflict does not matter. However, if the goal is to target and incentivize either group effort or individual effort and mitigate the other, the results from this experiment provide useful insight. For example, if group efforts are considered productive and individual efforts are considered wasteful, an increasing strength function would be preferred. By incentivizing higher group contributions through the promise of higher strength in internal competition, the principal actually discourages in-fighting without sacrificing collective effort. However, if only individual efforts are productive, then a decreasing strength function would be preferred to disincentivize group effort and heighten individual competition. Ultimately, the results of this paper suggest that an optimal organizational or industrial policy may depend on whether collective or individual effort is considered productive versus wasteful. If the relationship between contribution to group effort and strength is purely an artifact of a particular environment and not up to the discretion of the principal, then the results still provide descriptive information about how efforts are expended in different real-world settings.

For the next main set of results, I focus on the individual stage and explore how behavior varies across treatments with different relationships between the group stage and individual stage.[24]

[22] Wilcoxon rank-sum tests: INC vs. DEC, $p = 0.94$; INC vs. SYM, $p = 0.24$; INC vs. $RAND$, $p = 1.00$; DEC vs. SYM, $p = 0.132$; DEC vs. $RAND$, $p = 0.82$; SYM vs. $RAND$, $p = 0.09$

[23] Wald tests comparing constants in linear regression to predictions: INC, $p = 0.46$; DEC, $p = 0.14$; SYM, $p = 0.008$

[24] In tables 9 and 10 in Appendix B, I analyze the effects of previous round history on group stage and individual stage contributions. I include lags for previous round contributions and outcomes. I find that group stage contributions increase significantly in last-round group total contributions for all treatments and in winning the prize only for **Increasing**. Individual stage contributions also increase significantly in last round contributions in all treatments and in winning the prize in **Increasing**. Individual stage contributions decrease if the group won in the last round but the individual didn't win the prize.

Table 3: Individual Stage Contributions and Controls, Last 10 Rounds

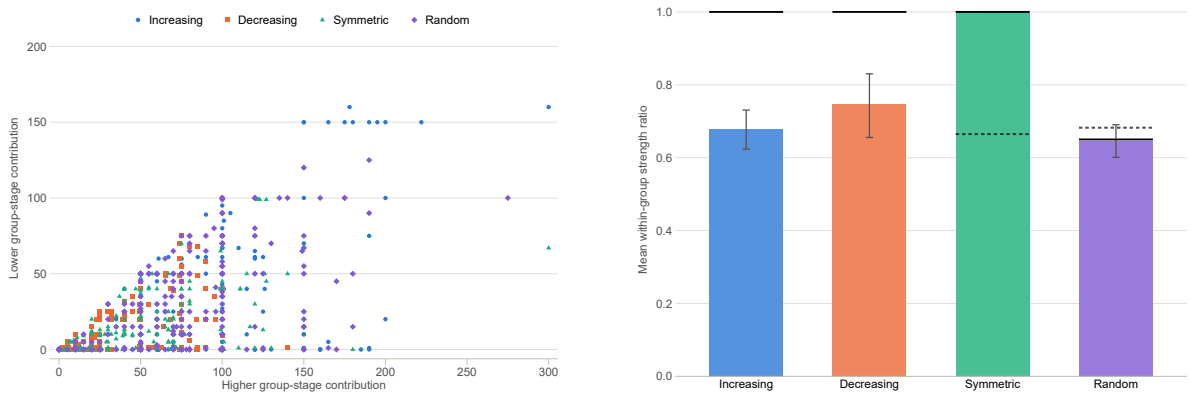
	(1)	(2)	(3)	(4)	(5)	(6)
Increasing	-4.829	-6.350	-4.450	-8.634	-3.576	-51.697
	[-27.091,17.432]	[-27.668,14.968]	[-26.993,18.092]	[-31.032,13.764]	[-28.697,21.545]	[-131.371,27.977]
Decreasing	34.871*	41.201*	33.294*	34.380*	35.895*	75.878
	[2.631,67.111]	[8.045,74.357]	[0.874,65.713]	[1.146,67.614]	[1.553,70.237]	[-35.362,187.118]
Symmetric	48.246***	51.766***	47.369***			
	[23.755,72.737]	[25.662,77.869]	[21.859,72.879]			
Group Stage contribution		0.183	-0.046	0.188	0.185	0.147
		[-0.035,0.402]	[-0.275,0.184]	[-0.046,0.422]	[-0.039,0.409]	[-0.059,0.352]
Group probability of winning			96.509***			
			[64.207,128.812]			
Strength Ratio				69.821*	331.966*	57.123
				[17.330,122.312]	[6.634,657.297]	[-24.062,138.307]
Strength Ratio Squared					-199.366	
					[-442.394,43.663]	
Increasing × Strength Ratio						64.599
						[-45.117,174.315]
Decreasing × Strength Ratio						-55.820
						[-193.659,82.019]
Constant	125.856***	116.325***	79.976***	70.944**	-7.192	81.314**
	[108.546,143.166]	[94.624,138.027]	[54.862,105.091]	[25.531,116.356]	[-116.688,102.304]	[26.633,135.995]
Observations	1920	1920	1920	1440	1440	1440
Clusters	24	24	24	18	18	18

Notes: Individual stage contributions regressed on treatment dummies and controls. Random treatment omitted. Symmetric treatment omitted in specifications (4)-(6). Strength ratio equals the lower strength in the group divided by the higher strength in the group. Data from last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

4.4 Behavior in the Individual Stage

In this section, I more closely investigate differences in contributions across treatments in the individual stage. Recall Hypothesis 4 which predicts that the strength ratio between contestants fully determines equilibrium contributions, and that how strengths are assigned should not affect behavior. In the analyses that follow, I test whether individuals respond differently to asymmetry depending on how asymmetry arises.[25]

[25]Recall that in my design, I use a "strategy method" approach. I withhold the outcome of the group stage until the end of each round, so all groups participate in the individual stage. However, the individual stage only affects payoffs for the group that wins. After submitting group stage contributions, subjects do see the probability that their group wins. There is a possibility that this could influence behavior in the individual stage. In figure 12 in Appendix B, I plot a positive relationship between the sum of individual stage contributions and the group probability of winning for each treatment. Individuals internalize the likelihood that the individual stage will affect their round payoffs and behave more competitively. This is consistent and similar across all treatments. I rerun the results of the following section on only the groups that actually win, only the groups that have a probability of winning greater than 0.5, and only the groups that have a probability of winning greater than 0.33. While precision and hypothesis test results change occasionally compared to studying the full sample, most of the qualitative results remain the same albeit with a higher level of individual stage contributions across the board. Therefore, I utilize the full sample for the following analyses.



(a) Group Coordination

(b) Strength Ratio

Figure 5: Group Coordination and Asymmetric Strengths, Last 10 Rounds

Notes: Panel (a): Lower group stage contribution over higher group stage contribution for each group by treatment. Each point represents one group. Data from last 10 rounds. Panel (b): Mean strength ratio by treatment. Black lines show theoretical predictions. Dashed lines show mean strength ratio in exogenous strength treatments if strength were endogenized. Data from last 10 rounds. Error bars represent 95% wild cluster bootstrap confidence intervals.

4.4.1 Within-Group Coordination and Asymmetry

Asymmetry in the individual stage arises when group mates select different group stage contributions or are randomly assigned different strengths. I now analyze the extent to which group members coordinate contributions in the group stage. Figure 5a shows a scatter plot in which each dot represents one group of two contestants and captures the ratio of the higher group stage contribution to the lower group stage contribution within a given group. Points along the diagonal represent perfect coordination in which both group members choose the same contribution level. Points along the horizontal axis represent perfect free-riding in which one group member contributes something positive while the other contributes 0. Points in between represent partial free-riding in which both group members contribute something positive but one out-contributes the other. In the treatments with endogenous strength, the theory predicts perfect coordination. In **Symmetric**, many equilibria exist involving asymmetric contributions. The scatter plot reveals that, while some perfect coordination and perfect free-riding exists, most groups involve group members choosing different positive group stage contributions. The same pattern arises regardless of treatment.

Figure 5b shows how differences in group stage contributions translate to individual stage asymmetry. I plot the mean within-group strength ratio for the individual stage in the last 10 rounds. Recall that strength ratio is the ratio of the group's minimum strength to the group's maximum strength. Mechanically, in **Symmetric**, since both group members are assigned a strength of 1 in the individual stage, strength ratio equals 1.

Theory predicts equal strengths in the endogenous strength treatments. I instead observe that in both endogenous strength treatments, the weaker group member is about 2/3 as strong as the stronger group member upon entering the individual stage. This corresponds with the average draw of strengths in **Random**. The strength ratio is slightly higher on average in **Decreasing** compared to **Increasing** and **Random**, but the difference is only significant when

compared to **Random**.[26] In appendix B, I plot histograms of strength ratios by treatment which reveal slightly more mass at lower strength ratios in **Increasing** even though there is no difference on average.

In figure 5b, I also plot a dashed line for each exogenous strength treatment representing the strength ratio that would've resulted if strength were endogenous. This again illustrates the similarity of the distribution of group stage contributions within groups across all four treatments.

4.4.2 Asymmetric Contests

Once subjects arrive in the individual stage and face an asymmetric contest, it should not matter how they got there. In theory, a subject who is twice as strong as her group mate shouldn't behave differently whether she is strong due to high group stage effort, low group stage effort, or a random draw. Any differences in effort should be explained only by differences in the degree of asymmetry between contestants within a group. Unexplained variation must be due to some behavioral biases regarding how asymmetry arises.[27]

I now test the robustness of the treatment differences in individual stage contributions to various regression specifications. Table 3 shows the results for six different specifications regressing individual stage contributions on treatment dummies and various control variables. In all specifications, **Random** is the excluded dummy variable.

The first specification is the naive model without any controls. Comparing mean individual stage contributions via regression analysis yields the same conclusions as the rank-sum tests.[28] The second specification controls for group stage contribution. Treatment differences remain robust, suggesting that a pure wealth effect isn't driving differences[29]

The remaining specifications control for strength ratio as a measure of asymmetry between contestants. These specifications drop data from **Symmetric** and focuses only on asymmetric contests. Strength ratio as defined above is calculated for each group as the minimum strength divided by the maximum strength. The strength ratio isolates the degree of asymmetry in the contest and ignores which player is stronger.[30] The final specification includes interaction

[26] Wilcoxon rank-sum test: INC vs. DEC, $p = 0.18$; INC vs. $RAND$, $p = 0.31$; DEC vs. $RAND$, $p = 0.03$

[27] Strictly speaking, in the endogenous strength treatments, there is intentionally and mechanically not random variation in strengths or measures of asymmetry. Subjects in these treatments certainly select into their strengths. However, once the individual stage arises, group stage behavior is sunk, and the realized strengths should in theory be treated no differently than if they were randomly assigned. Any differences that arise suggest some selection effects or other behavioral effects of endogenously assigning strengths. These biases are exactly what the following regression analyses seek to explore.

[28] Wald tests comparing constants in linear regression: INC vs DEC, $p = 0.03$; INC vs SYM, $p = 0.002$; INC vs. $RAND$, $p = 0.68$; DEC vs. SYM, $p = 0.43$; DEC vs. $RAND$, $p = 0.07$; SYM vs. $RAND$, $p = 0.005$

[29] In table 11, I regress individual stage contributions on group stage contributions interacted with treatment dummies. Individual stage contributions increase slightly in group stage contributions for every treatment except **Decreasing**, but the effects are significantly different from null only for **Increasing** (Wald tests comparing constants in linear regression to predictions: INC, $p = 0.09$; DEC, $p = 0.51$; SYM, $p = 0.99$; $RAND$, $p = 0.30$). There are also no significant differences in the effects between treatments(Wald tests comparing constants in linear regression: INC vs. DEC, $p = 0.18$; INC vs. SYM, $p = 0.28$; INC vs. $RAND$, $p = 0.37$; DEC vs. SYM, $p = 0.51$; DEC vs. $RAND$, $p = 0.29$; SYM vs $RAND$, $p = 0.68$). Combining these results with the results of specification 2 in table 3 above suggests that mere differences in group stage contributions don't drive the differences in individual stage contributions.

[30] Here, I intentionally separate the effects of whether a player is strong or weak from the effects of the degree of asymmetry. In the following subsections, I consider each one in turn. In tables 16 and 17, I offer an alternative

terms between treatment dummies and strength ratio to study potential slope differences across treatments.

Results are robust to controlling for strength ratio, except the difference between individual stage contributions in **Decreasing** and **Random** becomes insignificant in specifications 5 and 6.[31]

Result 4a (Strength assignment)**.** *Contrary to theory and Hypothesis 4, individual stage contributions vary across treatment for a given strength ratio.*

All in all, differences in individual stage contributions across treatments remain robust to controlling for different measures of group stage behavior or individual stage asymmetry. This warrants further investigation into what drives the treatment differences we observe. I now further investigate how asymmetry differently affects contributions across treatments.

4.4.3 Strength Ratio

Recall that theory predicts that the equilibrium level of contributions decreases as a contest becomes more asymmetric. To study this and compare across treatments, I regress individual stage contributions on treatment dummies interacted with strength ratio in specification 6 of table 3.

I compare coefficients on strength ratio to 0,[32] and I compare across treatments.[33] For both **Increasing** and **Random**, a higher strength ratio (greater symmetry) is positively correlated with individual stage contributions. This matches the theoretical prediction that greater symmetry induces more fierce competition. Contrary to theory, in **Decreasing** individual stage contributions decrease in strength ratio. The difference in the effect of strength ratio is significant when comparing the two endogenous strength treatments.

These results provide further evidence that individuals in asymmetric situations react differently depending on how asymmetry arises. In both endogenous strength treatments in my experiment, a gap in strengths arises due to differences in contributions to group effort. When contributing more to the group earns high strength, the greater the asymmetry, the less aggressive the competition. Again, it seems competitors in this case accept that the higher group contributor is entitled in some way to the higher likelihood of winning the prize. I see similar results when strengths are randomly assigned. In either case, players are more inclined to accept the strength distribution and compete less fiercely as a result. On the other hand, when the shirker in the group attains higher strength, players behave more competitively the wider the strength gap becomes. In this case, players don't accept that the stronger player has earned the right to the prize and still compete aggressively. The more severe the asymmetry, the more aggressive the competition. Once again, the main contextual factor which affects behavior is how the advantaged position is earned.

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approach to try to isolate the effects of asymmetry and position in one regression. I regress individual stage contributions on strength ratio and a dummy for whether the player is stronger or weaker and report the slope coefficients on strength ratio for each treatment-type combination. In doing so, I allow the effects of asymmetry to differ between stronger and weaker players and across treatments.

[31] Wald tests comparing constants in linear regression: DEC vs. $RAND$, $p = 0.13$ in specification 5; DEC vs. $RAND$, $p = 0.13$ in specification 6

[32] Wald tests comparing constants in linear regression to predictions: INC, $p = 0.03$; DEC, $p = 0.98$; $RAND$, $p = 0.21$

[33] Wald tests comparing constants in linear regression to predictions: INC vs. DEC, $p = 0.09$; INC vs. $RAND$, $p = 0.26$; DEC vs. $RAND$, $p = 0.41$

Table 4: Individual Stage Contributions by Treatment and Type, Last 10 Rounds

Type	Increasing	Decreasing	Symmetric	Random
Equal	146.74	133.60	174.10	126.14
	[71.29,178.82]	[96.15,182.28]	[151.58,188.34]	[84.37,159.25]
Strong	133.20	181.69	.	139.19
	[105.95,168.52]	[118.02,236.77]		[115.63,165.16]
Weak	99.78	154.78	.	112.46
	[73.93,131.66]	[111.80,206.29]		[87.90,140.90]

Notes: Mean individual stage contributions by treatment and type. Estimates from regressing individual stage contribution on treatment dummy and type dummy. Data from last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.

Result 4b (Effect of degree of asymmetry)**.** *Greater asymmetry affects competition differently depending on how asymmetry arises. When a strong position is earned through high effort or randomly assigned, greater asymmetry dampens competition. On the contrary, when a strong position is earned through shirking, greater asymmetry heightens competition.*

4.4.4 Strong vs. Weak Types

Next, I set aside the degree of strength asymmetry and focus on the effect of being stronger or weaker relative to group mate.[34] I compare individual stage contributions and percent deviations from stage Nash equilibrium across treatment and type (equal, strong, or weak). Table 4 shows mean contributions by treatment and type and table 14 in Appendix B shows mean percent deviation. In these tables, equality is defined broadly as being within a range based on the absolute strength values in each treatment (within 0.2 in **Increasing** and **Random** and within 0.02 in **Decreasing**). In tables 13 and 15 in Appendix B, I consider only exact equality, and the qualitative results do not change.

To test whether overinvestment occurs, I analyze where percent deviation from Nash equilibrium differs from 0. Contrary to previous research, I find that not all types exhibit overinvestment. Equal types in both endogenous strength treatments and the weak types from **Increasing** and **Random** actually underinvest on average. Strong types and subjects in **Symmetric** (equal by default) don't deviate from Nash equilibrium on average or slightly overinvest. It appears the lack of overinvestment we observe is driven by the weak and equal types. Deviations are statistically significant for **Symmetric** ($p = 0.04$) and the weak types in **Increasing** ($p = 0.07$) and **Random** ($p = 0.08$).

In my experiment, I have symmetric contests that arise exogenously and endogenously, so I compare behavior in asymmetric contests to each in turn. Contributions by both strong and weak types in **Increasing** and **Random** are significantly lower than contributions in **Symmetric**,

[34] I can compare this analysis to previous research on asymmetric contests (particularly Fonseca (2009)) and explore whether results differ when asymmetry arises endogenously. Fonseca (2009) compares symmetric and asymmetric contests where strength is randomly assigned for each player. Each treatment involves either a symmetric or an asymmetric format, and subjects within treatments are categorized as being equal, weak, or strong. The author draws three main conclusions from the experimental data. Firstly, subjects in symmetric contests and both types in asymmetric contests all overinvest on average. Secondly, both types in asymmetric contests invest less on average than subjects in symmetric contests. Thirdly, both types in asymmetric contests invest about the same level on average.

while neither type in **Decreasing** contributes differently.[35] Contributions in an endogenously symmetric contest are significantly lower in both endogenous strength treatments on average compared to contributions in an exogenously symmetric contest.[36] Therefore, the results differ when comparing contributions in endogenously asymmetric contests to endogenously symmetric contests as a benchmark. In **Increasing**, there is no longer any significant difference for either type, while in **Decreasing**, contributions by strong and weak types are actually *higher* and significantly so for the strong types.[37]

Regardless of which symmetric contest benchmark I use (exogenously symmetric or endogenously symmetric), contrary to theoretical predictions and previous experimental findings, I do not find that both strong and weak types in asymmetric contests always contribute less on average than those in symmetric contests. In particular, both types in **Decreasing** compete similarly to if they were symmetric.

Comparing types within treatments, the strong player contributes slightly more on average than the weak player in all three treatments involving asymmetry. The difference is statistically significant in **Random**.[38] Equality across types is the prediction of theory and the empirical observation of Fonseca (2009).

I next compare within types across treatments. Previous research on asymmetric contests has been unable to perform this analysis because it holds fixed the method for assigning types. I observe that within strong types, there are no differences between **Increasing** and **Random**. However, strong types contribute more in **Decreasing** on average compared to both **Increasing** and **Random**.[39] Similarly, for weak types, there are no differences between **Increasing** and **Random**, but weak types contribute more in **Decreasing** on average compared to both **Increasing** and **Random**, and these differences are statistically significant.[40]

Across the board, strong and weak types in **Decreasing** contribute more on average than their counterparts in other treatments. When higher strength is earned through shirking as opposed to hard work or random assignment, it appears overall competitiveness remains high. On the other hand, competitiveness dampens when a high position is earned through hard work or given exogenously.

Interestingly, equal types in **Decreasing** do not contribute significantly more on average than their counterparts in **Increasing**, and both equal types in endogenous strength treatments contribute significantly less than subjects in **Symmetric**[41]. When individuals are evenly matched due to endogenous coordination, they behave less competitively against each other on average compared to when they are evenly matched exogenously.

Finally, I compare across treatments the share of individual stage contributions made by the strong type. In figure 6, I plot the stronger player's share of the total individual stage

[35] Wald tests comparing constants in linear regression: INC_{strong} , $p = 0.03$; INC_{weak} , $p = 0.004$; DEC_{strong} , $p = 0.74$; DEC_{weak} , $p = 0.33$; $RAND_{strong}$, $p = 0.02$; $RAND_{weak}$, $p = 0.002$

[36] Wald tests comparing constants in linear regression: INC_{equal} , $p = 0.03$; DEC_{equal} , $p = 0.03$

[37] Wald tests comparing constants in linear regression: INC_{strong} , $p = 0.98$; INC_{weak} , $p = 0.34$; DEC_{strong} , $p = 0.08$; DEC_{weak} , $p = 0.24$

[38] Wald tests comparing constants in linear regression: INC_{strong} vs. INC_{weak}, $p = 0.21$; DEC_{strong} vs. DEC_{weak}, $p = 0.28$; $RAND_{strong}$ vs. $RAND_{weak}$, $p = 0.01$

[39] Wald tests comparing constants in linear regression: INC_{strong} vs. DEC_{strong}, $p = 0.12$; INC_{strong} vs. $RAND_{strong}$, $p = 0.79$; DEC_{strong} vs. $RAND_{strong}$, $p = 0.14$

[40] Wald tests comparing constants in linear regression: INC_{weak} vs. DEC_{weak}, $p = 0.03$; INC_{weak} vs. $RAND_{weak}$, $p = 0.65$; DEC_{weak} vs. $RAND_{weak}$, $p = 0.04$

[41] Wald tests comparing constants in linear regression: INC_{equal} vs. DEC_{equal}, $p = 0.72$; INC_{equal} vs. SYM, $p = 0.03$; DEC_{equal} vs. SYM, $p = 0.03$

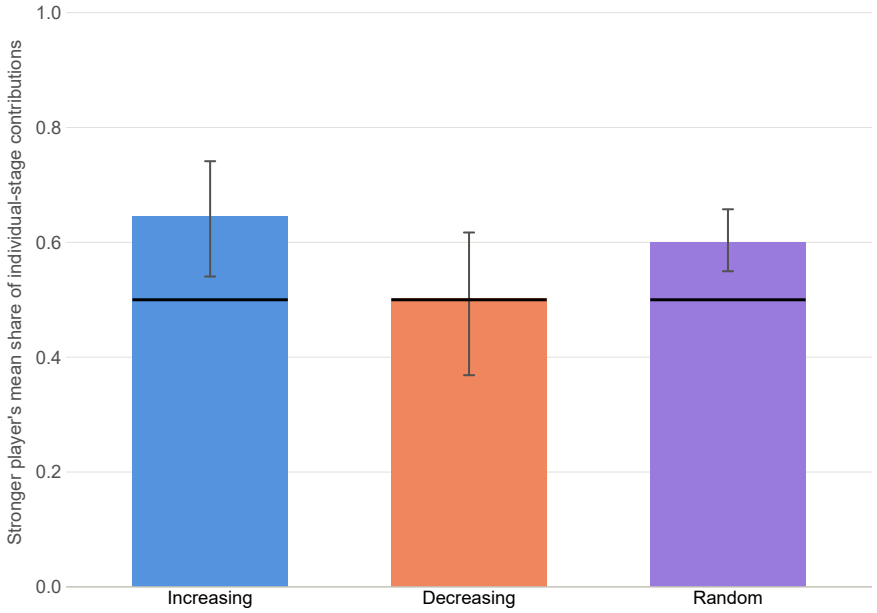


Figure 6: Share of Individual Stage Contributions, Last 10 Rounds

Notes: The stronger player's share of the total individual stage contributions within a group for each treatment in which asymmetric strengths can exist. Black lines show theoretical predictions. Data from last 10 rounds. Error bars represent 95% wild cluster bootstrap confidence intervals.

contributions in only asymmetric contests. Theory predicts equal individual stage contributions for both contestants even if asymmetry arises and no differences between treatments.

In **Decreasing**, the stronger and weaker player both contribute about an equal share of the total individual stage contributions. However, in **Increasing** and **Random**, the stronger player behaves relatively more aggressively, contributing about 64% and 60%, respectively, of the total individual stage contributions on average. In both treatments, the stronger player contributes a share which is significantly greater than an even split.[42] The stronger player's share is significantly higher in **Increasing** compared to **Decreasing**.[43] In table 18, I also regress the stronger player's share of individual stage contributions on treatment and strength ratio. I compare the slope coefficients on strength ratio to 0,[44] and I compare across treatments.[45]

I find a significant difference in the effect of strength ratio on the stronger player's share of contributions between the two endogenous strength treatments and between **Increasing** and **Random**. When a subject earns a strong position in **Increasing**, greater asymmetry increases her share of the contributions. The effect is stronger than when a subject becomes strong exogenously as in **Random**. On the contrary, stronger players who receive a strong position by being outworked as in **Decreasing** actually submit a lower share of contributions as the strength gap widens.

[42] Wald tests comparing constants in linear regression: INC, $p = 0.04$; DEC, $p = 0.96$; $RAND$, $p = 0.01$

[43] Wilcoxon rank-sum test: INC vs. DEC, $p = 0.04$; INC vs. $RAND$, $p = 0.24$; DEC vs. $RAND$, $p = 0.18$

[44] Wald tests comparing constants in linear regression: INC, $p = 0.009$; DEC, $p = 0.84$; $RAND$, $p = 0.09$

[45] Wald tests comparing constants in linear regression: INC vs. DEC, $p = 0.09$; INC vs. $RAND$, $p = 0.006$; DEC vs. $RAND$, $p = 0.63$

The results on the asymmetric contests in this experiment suggest that both strong and weak types in an asymmetric contest behave differently depending on how asymmetry arises.

Result 4c (Strong vs. weak)**.** *Weak types contribute significantly more on average when strength is decreasing versus when it is increasing. The share of individual stage contributions expended by the stronger player is greater when strength is increasing versus when it is decreasing.*

4.4.5 Discussion

My results resemble a nuanced expression of the entitlement effect first observed in dictator and ultimatum games (Hoffman et al., 1994; Cappelen et al., 2007; Oxoby & Spraggon, 2008). Typically, individuals behave less generously when they earn an advantaged position compared to when it is given. In my competitive setting, I find that people also seem to care about *how* an advantaged position is earned. When the strong position is earned through high group contribution, both players appear to accept the outcome and reduce competition. This is especially true for weak players. Players also seem to accept the strength distribution as fair and dampen competition when strengths are randomly assigned. However, when the strong position is earned through a lack of group contribution, then competition ramps up. It appears neither player accepts that the stronger group member is entitled to the prize. Contrary to previous research, individuals don't care as much about the distinction between earned versus given advantage, but they do respond to differences in how advantage is earned. The difference in how asymmetry arises also affects how stronger versus weaker players compete. When a subject earns an advantage by outworking her group mate, she competes relatively aggressively in the individual stage. When a subject receives an advantage after being outworked, she matches her weaker group mate's competitiveness in the individual stage. This provides more evidence that subjects accept the stronger player's claim to the prize when the strong position is earned through high group contribution or randomly assigned but not when the strong position is earned through shirking.

My results also shed light on discouragement effects in asymmetric contests. The discouragement effect describes how contributions in a contest to fall when contestants are asymmetric. In my experiment, the discouragement effect only arises when strength increases in group stage contributions. It appears the discouragement effect may be sensitive to the particular source of asymmetry.

5 Conclusion

In this paper, I study situations in which individuals coordinate and compete for prizes. Such situations commonly arise in the real world in organizations, sports, markets and political economy. I focus on the link between an individual's contribution to a group's collective effort and her position in competition within the group. To study this topic, I develop a two-stage model of intergroup and intragroup competition. In the group stage, two groups of two agents each compete in a group Tullock contest for the right to advance to the next stage. Group members coordinate and pool their contributions to increase their likelihood of advancing. The group that wins moves on to the individual stage in which group members compete against one another for an indivisible prize. Contributions in the individual stage are multiplied by strength. Strength may be exogenously or endogenously determined and symmetric or asymmetric. Endogenous

strength depends on an individual's contributions in the group stage. In one case, higher group stage contributions translate to higher strength, while in another case higher group stage contributions translate to lower strength. I solve the model for the different cases of strength relationships and develop testable hypotheses regarding equilibrium contributions in each stage and comparative statics.

I then conduct a laboratory experiment in which subjects play the two-stage game for 20 repeated rounds. I analyze the data on contributions in each stage and find two main sets of results. First, within each stage, contributions differ depending on the method for assigning strengths, while total contributions over both stages remain constant. Consequently, the optimal method for assigning individual strengths from society or a principal's perspective depends on whether collective or individual effort is considered productive versus wasteful.

Second, in analyzing the individual stage in my experiment, I am able to study asymmetric contests with different origins of asymmetry. Contrary to theoretical predictions, I find that individuals competing in asymmetric Tullock contests are indeed sensitive to how asymmetry arises. In particular, the distribution of strengths isn't the only factor that affects contribution decisions. Individuals also care about whether strength depends on group contributions and how exactly it is determined. Specifically, I find that individuals behave more competitively when an advantaged position of high strength comes from low group contributions compared to when an advantage is randomly assigned or earned through high group contributions. Within firms and other organizations, unevenly matched agents frequently interact and often compete. According to my results, managers and policymakers overseeing such situations should pay close attention to the origins of asymmetry and adapt their strategies accordingly to optimize the effectiveness of their agents.

This paper is the first to theoretically and experimentally study the relationship between group contributions and individual strength in competition between and within groups. The results advance the vast literature on group and individual contests. Even more importantly, this research offers new insights into the behavioral biases individuals possess towards the assignment of property rights or other advantages. This paper also provides new insights into how groups of individuals coordinate and compete within organizations. Future research should explore optimal policy responses to the behavioral patterns I recognize.

References

Altmann, S., Falk, A., & Wibral, M. (2012). Promotions and incentives: The case of multistage elimination tournaments. *Journal of Labor Economics*, *30*(1), 149–174.

Baik, K. H. (1994). Effort levels in contests with two asymmetric players. *Southern Economic Journal*, 367–378.

Bhattacharya, P., & Rampal, J. (2024). Contests within and between groups: Theory and experiment. *Games and Economic Behavior*, *145*, 467–492.

Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *The review of economics and statistics*, *90*(3), 414–427.

Cappelen, A. W., Hole, A. D., Sørensen, E. Ø., & Tungodden, B. (2007). The pluralism of fairness ideals: An experimental approach. *American Economic Review*, *97*(3), 818–827.

Cherry, T. L., Frykblom, P., & Shogren, J. F. (2002). Hardnose the dictator. *American economic review*, *92*(4), 1218–1221.

Choi, J. P., Chowdhury, S. M., & Kim, J. (2016). Group contests with internal conflict and power asymmetry. *The Scandinavian Journal of Economics*, *118*(4), 816–840.

Chowdhury, S. M., Esteve-González, P., & Mukherjee, A. (2023). Heterogeneity, leveling the playing field, and affirmative action in contests. *Southern Economic Journal*, *89*(3), 924–974.

Cross, R., Rebele, R., & Grant, A. (2016). Collaborative overload (harvard business review). *Online verfügbar unter https://hbr. org/2016/01/collaborative-overload, zuletzt geprüft am*, *23*, 2019.

Dechenaux, E., Kovenock, D., & Sheremeta, R. M. (2015). A survey of experimental research on contests, all-pay auctions and tournaments. *Experimental Economics*, *18*(4), 609–669.

Fischbacher, U. (2007). z-tree: Zurich toolbox for ready-made economic experiments. *Experimental economics*, *10*(2), 171–178.

Fonseca, M. A. (2009). An experimental investigation of asymmetric contests. *International Journal of Industrial Organization*, *27*(5), 582–591.

Franke, J., Kanzow, C., Leininger, W., & Schwartz, A. (2013). Effort maximization in asymmetric contest games with heterogeneous contestants. *Economic Theory*, *52*(2), 589–630.

Fu, Q., & Lu, J. (2012). The optimal multi-stage contest. *Economic Theory*, *51*(2), 351–382.

Gradstein, M., & Konrad, K. A. (1999). Orchestrating rent seeking contests. *The Economic Journal*, *109*(458), 536–545.

Greiner, B. (2015). Subject pool recruitment procedures: organizing experiments with orsee. *Journal of the Economic Science Association*, *1*(1), 114–125.

Hoffman, E., McCabe, K., Shachat, K., & Smith, V. (1994). Preferences, property rights, and anonymity in bargaining games. *Games and Economic behavior*, *7*(3), 346–380.

Katz, E., Nitzan, S., & Rosenberg, J. (1990). Rent-seeking for pure public goods. *Public Choice*, *65*(1), 49–60.

Katz, E., & Tokatlidu, J. (1996). Group competition for rents. *European Journal of Political Economy*, *12*(4), 599–607.

Kirkegaard, R. (2012). Favoritism in asymmetric contests: Head starts and handicaps. *Games and Economic Behavior*, *76*(1), 226–248.

Mealem, Y., & Nitzan, S. (2016). Discrimination in contests: a survey. *Review of Economic Design*, *20*(2), 145–172.

Münster, J. (2007). Simultaneous inter-and intra-group conflicts. *Economic Theory*, *32*(2), 333–352.

Münster, J. (2009). Group contest success functions. *Economic Theory*, *41*(2), 345–357.

Oxoby, R. J., & Spraggon, J. (2008). Mine and yours: Property rights in dictator games. *Journal of Economic Behavior & Organization*, *65*(3-4), 703–713.

Rogerson, W. (1982). The social costs of monopoly and regulation: A game-theoretic analysis. *Bell Journal of Economics*, *13*(2), 391–401.

Ryvkin, D. (2011). Fatigue in dynamic tournaments. *Journal of Economics & Management Strategy*, *20*(4), 1011–1041.

Schmitt, P., Shupp, R., Swope, K., & Cadigan, J. (2004). Multi-period rent-seeking contests with carryover: Theory and experimental evidence. *Economics of Governance*, *5*(3), 187–211.

Schotter, A., & Weigelt, K. (1992). Asymmetric tournaments, equal opportunity laws, and affirmative action: Some experimental results. *The Quarterly Journal of Economics*, *107*(2), 511–539.

Sheremeta, R. M. (2010a). Expenditures and information disclosure in two-stage political contests. *Journal of Conflict Resolution*, *54*(5), 771–798.

Sheremeta, R. M. (2010b). Experimental comparison of multi-stage and one-stage contests. *Games and Economic Behavior*, *68*(2), 731–747.

Sheremeta, R. M. (2018). Behavior in group contests: A review of experimental research. *Journal of Economic Surveys*, *32*(3), 683–704.

Skaperdas, S. (1996). Contest success functions. *Economic theory*, *7*(2), 283–290.

Stein, W. E. (2002). Asymmetric rent-seeking with more than two contestants. *Public Choice*, *113*(3), 325–336.

Stein, W. E., & Rapoport, A. (2004). Asymmetric two-stage group rent-seeking: Comparison of two contest structures. *Public Choice*, *118*(1), 151–167.

Tsoulouhas, T., Knoeber, C. R., & Agrawal, A. (2007). Contests to become ceo: incentives, selection and handicaps. *Economic Theory*, *30*(2), 195–221.

Tullock, G. (1967). The welfare costs of tariffs, monopolies, and theft. *Economic Inquiry*, *5*(3), 224–232.

Tullock, G. (1980). Efficient rent seeking. In *Toward a theory of the rent-seeking society* (p. 97-112). Texas A&M University Press.

Wärneryd, K. (1998). Distributional conflict and jurisdictional organization. *Journal of public economics*, *69*(3), 435–450.

A Additional Theoretical Results and Proofs

A.1 Individual stage equilibrium with binding budget constraints

This appendix characterizes the Nash equilibrium of the individual stage when players may be budget constrained. Suppress group indices for this analysis. Let remaining endowments after the group stage be

$$r_i \equiv w - x_i \geq 0, \qquad i \in \{1, 2\},$$

and let strengths be (θ_1, θ_2), where $\theta_i > 0$. In the individual stage, agents choose $y_i \in [0, r_i]$ to maximize

$$u_i(y_i, y_j) = \frac{\theta_i y_i}{\theta_i y_i + \theta_j y_j} V - y_i + r_i, \tag{13}$$

The unconstrained first-order condition for agent i given the contribution of agent j is

$$\frac{\partial u_i}{\partial y_i} = \frac{V \theta_i \theta_j y_j}{(\theta_i y_i + \theta_j y_j)^2} - 1 = 0,$$

and the unconstrained best response function is

$$y_i(y_j) = \frac{\sqrt{V \theta_i \theta_j y_j} - \theta_j y_j}{\theta_i}. \tag{14}$$

Imposing the budget constraint $y_i \leq r_i$ implies the constrained best response

$$BR_i(y_j) = \min \left\{ r_i,\ \max \left\{ y_i(y_j) \right\} \right\}. \tag{15}$$

If budgets do not bind for either player, the unique interior Nash equilibrium is the standard outcome characterized in Lemma 1:

$$y_1^* = y_2^* = \frac{V \theta_1 \theta_2}{(\theta_1 + \theta_2)^2}. \tag{16}$$

Suppose the budget constraint binds only for agent 1. Then the Nash equilibrium satisfies

$$y_1^* = r_1, \qquad y_2^* = \frac{\sqrt{V \theta_1 \theta_2 r_1} - \theta_1 r_1}{\theta_2}, \tag{17}$$

Suppose both agents are constrained. Then both constraints bind in equilibrium:

$$y_1^* = r_1, \qquad y_2^* = r_2. \tag{18}$$

Note that in our experiment, budget constraints never bind on the equilibrium path. However, budget constraints may bind off the equilibrium path. Deviating to a group stage contribution that makes budget constraints bind in the individual stage is unprofitable. Therefore, even if budget constraints can bind in the individual stage for certain group stage contributions, the main analysis of the subgame-perfect equilibria given my experimental parameters remains unchanged.

A.2 Proofs

Proof of Lemma 1

Proof. Assume for all (x_{1a}, x_{2a}) that arise in equilibrium, budget constraints do not bind in the individual stage:

$$w - x_{ia} \;\geq\; \frac{V\theta_{1a}\theta_{2a}}{(\theta_{1a}+\theta_{2a})^2} \quad \text{for } ia \in \{1a, 2a\},$$

and $f(x) > 0$ on $[0, w]$. The payoff function is strictly concave in y_{ia} for $y_{ja} > 0$, so first-order conditions are sufficient and yield the unique interior Nash equilibrium. The FOCs are

$$\frac{V\theta_{1a}\theta_{2a}y_{2a}}{(\theta_{1a}y_{1a}+\theta_{2a}y_{2a})^2} - 1 = 0 \qquad \frac{V\theta_{1a}\theta_{2a}y_{1a}}{(\theta_{1a}y_{1a}+\theta_{2a}y_{2a})^2} - 1 = 0$$

which implies $y^*_{1a} = y^*_{2a}$. Substitution gives the winning probabilities and continuation payoffs. □

Proof of Lemma 2

Proof. Consider agent $1a$. Differentiate π_{1a} with respect to x_{1a} and impose the first-order condition. The simplified FOC results in

$$2\frac{X_a}{X_a+X_b}\left(\frac{f(x_{1a})f(x_{2a})f'(x_{1a})}{(f(x_{1a})+f(x_{2a}))^3}\right) + \frac{X_b}{(X_a+X_b)^2}\frac{f(x_{1a})^2}{(f(x_{1a})+f(x_{2a}))^2} = 1 \tag{19}$$

A symmetric FOC exists for each agent. Consider a symmetric profile

$$x_{1a} = x_{2a} = x_{1b} = x_{2b} = x.$$

Then $X_a = X_b = 2x$ and $f(x_{1a}) = f(x_{2a}) = f(x_{1b}) = f(x_{2b}) = f(x)$. Within the winning group,

$$U_{1a} = U_{2a} = \frac{V}{4}.$$

Imposing symmetry and simplifying the above FOC further results in the symmetric equilibrium condition. □

Proof of Proposition 1

Proof. Because the individual stage equilibrium is unique and yields $u_{ig} = V/4$ for both agents in the winning group, stage-1 equilibrium reduces to Nash equilibrium of the group stage game with payoffs equal to (5).

Differentiating expected payoff with respect to x_{ig} yields

$$\frac{\partial \pi_{ig}}{\partial x_{ig}} = -1 + \frac{V}{4}\cdot\frac{\partial}{\partial x_{ig}}\left(\frac{x_{ig}+x_{jg}}{x_{ig}+x_{jg}+X_{-g}}\right) = -1 + \frac{V}{4}\cdot\frac{X_{-g}}{(X_g+X_{-g})^2}.$$

Since

$$\frac{\partial^2 \pi_{ig}}{\partial x_{ig}^2} = \frac{V}{4}\cdot X_{-g}\cdot\frac{\partial}{\partial x_{ig}}(X_g+X_{-g})^{-2} = -\frac{V}{2}\cdot\frac{X_{-g}}{(X_g+X_{-g})^3} < 0 \quad \text{whenever } X_{-g} > 0,$$

the objective is strictly concave in x_{ig} for $X_{-g} > 0$, so the first-order condition is sufficient for a unique best response.

An interior best response satisfies the following FOCs:

$$-1 + \frac{V}{4} \cdot \frac{X_b}{(X_a + X_b)^2} = 0 \qquad -1 + \frac{V}{4} \cdot \frac{X_a}{(X_a + X_b)^2} = 0$$

which implies $X_a = X_b$. An equilibrium can only pin down aggregate group stage contributions. Within each group g, any split (x_{1g}, x_{2g}) with $x_{1g}, x_{2g} \geq 0$ and $x_{1g} + x_{2g} = V/16$ constitutes mutual best responses given that the opposing group's total is also $V/16$.

The individual stage equilibrium follows from Lemma 1 and is unique. □

Proof of Proposition 2

Proof. By Lemma 1, the individual stage has a unique Nash equilibrium in which

$$y^*_{1g} = y^*_{2g} = \frac{V\theta_{1g}\theta_{2g}}{(\theta_{1g} + \theta_{2g})^2}, \qquad p^*_{ig} = \frac{\theta_{ig}}{\theta_{1g} + \theta_{2g}}, \qquad u_{ig} = \frac{V\theta^2_{ig}}{(\theta_{1g} + \theta_{2g})^2}.$$

In particular, under a symmetric group stage profile, $\theta_{1g} = \theta_{2g} = f(x)$ so $y^*_{1g} = y^*_{2g} = V/4$ and $u_{1g} = u_{2g} = V/4$.

Substituting the continuation values into the group stage payoff yields

$$\pi_{ig}(x_{ig}, x_{jg}, X_{-g}) = w - x_{ig} + p_g(X_g, X_{-g})\, U_i\big(f(x_{ig}), f(x_{jg})\big).$$

By Lemma 2, any symmetric interior equilibrium $x^* > 0$ must satisfy

$$1 = \frac{V}{32x^*} + \frac{V}{8}\frac{f'(x^*)}{f(x^*)}.$$

Under $f(x) = (x + \alpha)^\gamma$, this condition becomes the equality (6).

Define the symmetric first-order condition

$$FOC(x) \equiv \frac{V}{32x} + \frac{V\gamma}{8(x + \alpha)} - 1, \qquad x \in (0, w].$$

Then FOC is continuous on $(0, w]$ and $\lim_{x \downarrow 0} FOC(x) = +\infty$. Moreover,

$$FOC'(x) = -\frac{V}{32x^2} - \frac{V\gamma}{8(x + \alpha)^2} < 0,$$

so FOC is strictly decreasing. Hence $FOC(x) = 0$ has at most one solution, and if $FOC(w) \leq 0$, existence and uniqueness of $x^* \in (0, w]$ follow by the Intermediate Value Theorem.

Individual stage equilibrium is uniquely pinned down by Lemma 1 for every realized vector of strengths. Uniqueness holds because the symmetric group stage equilibrium is unique and the individual stage equilibrium is unique given strengths. □

Proof of Corollary 1

Proof. By Lemma 1, conditional on group g winning the group stage, agent ig's continuation payoff is

$$u_{ig}(x_{ig}, x_{jg}) = V \frac{f(x_{ig})^2}{\big(f(x_{ig}) + f(x_{jg})\big)^2} - x_{ig} + w. \tag{20}$$

So the reduced two-stage payoff is

$$\pi_{ig} = w - x_{ig} + p_g\, u_{ig}(x_{ig}, x_{jg}).$$

The first-order condition for agent ig is

$$-1 + p_g' u_{ig}(x_{ig}, x_{jg}) + p_g \frac{\partial u_{ig}}{\partial x_{ig}}(x_{ig}, x_{jg}) = 0. \tag{21}$$

Towards a contradiction, suppose that there exists an interior equilibrium with $x_{1g} \neq x_{2g}$. Without loss of generality, let $x_{1g} > x_{2g}$. Subtracting the FOCs in (21) for agents $1g$ and $2g$ yields

$$p_g'\big(u_{1g} - u_{2g}\big) + p_g \left(\frac{\partial u_{1g}}{\partial x_{1g}} - \frac{\partial u_{2g}}{\partial x_{2g}} \right) = 0. \tag{22}$$

Differentiating equation (20) gives

$$\frac{\partial u_{ig}}{\partial x_{ig}} = 2V \cdot \frac{f(x_{ig}) f(x_{jg})}{\big(f(x_{ig}) + f(x_{jg})\big)^3} f'(x_{ig}). \tag{23}$$

Substituting into equation (22) and simplifying yields

$$\frac{p_g'}{p_g} = -2 \cdot \frac{f(x_{1g}) f(x_{2g})}{\big(f(x_{1g}) + f(x_{2g})\big)^2} \cdot \frac{f'(x_{1g})/f(x_{1g}) - f'(x_{2g})/f(x_{2g})}{\frac{f(x_{1g})^2 - f(x_{2g})^2}{f(x_{1g}) f(x_{2g})}}. \tag{24}$$

First note that the left-hand side of equation (24) is strictly positive:

$$\frac{p_g'}{p_g} = \frac{X_{-g}}{X_g(X_g + X_{-g})} > 0.$$

Now consider the right-hand side. When $x_{1g} > x_{2g}$ and $\gamma \in \{\frac{1}{3}, -\frac{1}{3}\}$, we have that f is strictly monotone and $f'(x)/f(x) = \gamma/(x+3)$. Hence

$$\frac{f'(x_{1g})}{f(x_{1g})} - \frac{f'(x_{2g})}{f(x_{2g})} = \frac{\gamma}{x_{1g} + 3} - \frac{\gamma}{x_{2g} + 3},$$

which has the opposite sign of γ. Moreover,

$$f(x_{1g})^2 - f(x_{2g})^2$$

has the same sign as γ. It follows that the fraction in equation (24) is negative when $\gamma > 0$ and positive when $\gamma < 0$, implying that the entire right-hand side of equation (24) is strictly negative for both $\gamma = \pm 1/3$:

$$-2 \cdot \frac{f(x_{1g}) f(x_{2g})}{\big(f(x_{1g}) + f(x_{2g})\big)^2} \cdot \frac{f'(x_{1g})/f(x_{1g}) - f'(x_{2g})/f(x_{2g})}{\frac{f(x_{1g})^2 - f(x_{2g})^2}{f(x_{1g}) f(x_{2g})}} < 0.$$

This contradicts equation (24), since its left-hand side is positive. Therefore, no interior equilibrium can satisfy $x_{1g} > x_{2g}$. By symmetry of the argument, $x_{2g} > x_{1g}$ is also impossible. Hence $x_{1g} = x_{2g}$ in any interior equilibrium.

Since groups are symmetric, it must be that $X_a = X_b$ in equilibrium. If $X_a = X_b$ and $x_{1g} = x_{2g}$ for all g, then it must be that $x_{1a} = x_{2a} = x_{1b} = x_{2b} = x$. Therefore, any equilibrium must have symmetric group stage contributions. □

Proof of Remark 2

Proof. Consider two strengths $\theta' > \theta$. Suppose, toward a contradiction, that

$$x(\theta') \leq x(\theta). \tag{25}$$

In equilibrium, given realized strength, group stage contribution satisfies the first order condition in (11). Therefore, for type θ' and θ, the marginal benefit of increasing group stage contribution equals 1, $MB(\theta') = 1$ and $MB(\theta) = 1$. The marginal benefit strictly increases in strength and decreases in group stage contribution. Therefore, if $\theta' > \theta$ and $x(\theta') \leq x(\theta)$, then $MB(\theta') > MB(\theta) = 1$. But in equilibrium, $MB(\theta') = 1$, so we have a contradiction.

□

B Supplementary Results

Table 5: Summary Statistics, All Rounds

	Group Stage		Individual Stage	
Treatment	Mean	SPE	Mean	SPE
Increasing	68.41	42.09	125.05	150.00
	[42.02, 98.27]		[108.97, 151.44]	
Decreasing	25.37	4.19	147.18	150.00
	[11.64, 45.94]		[115.95, 174.20]	
Symmetric	42.63	18.75	162.79	150.00
	[34.42, 49.87]		[138.62, 175.00]	
Random	59.66	27.55	123.22	143.25
	[39.64, 82.27]		[99.50, 143.68]	

Notes: Mean contributions by stage and treatment with theoretical predictions. Data from all rounds. 95% wild cluster bootstrap confidence intervals in brackets.

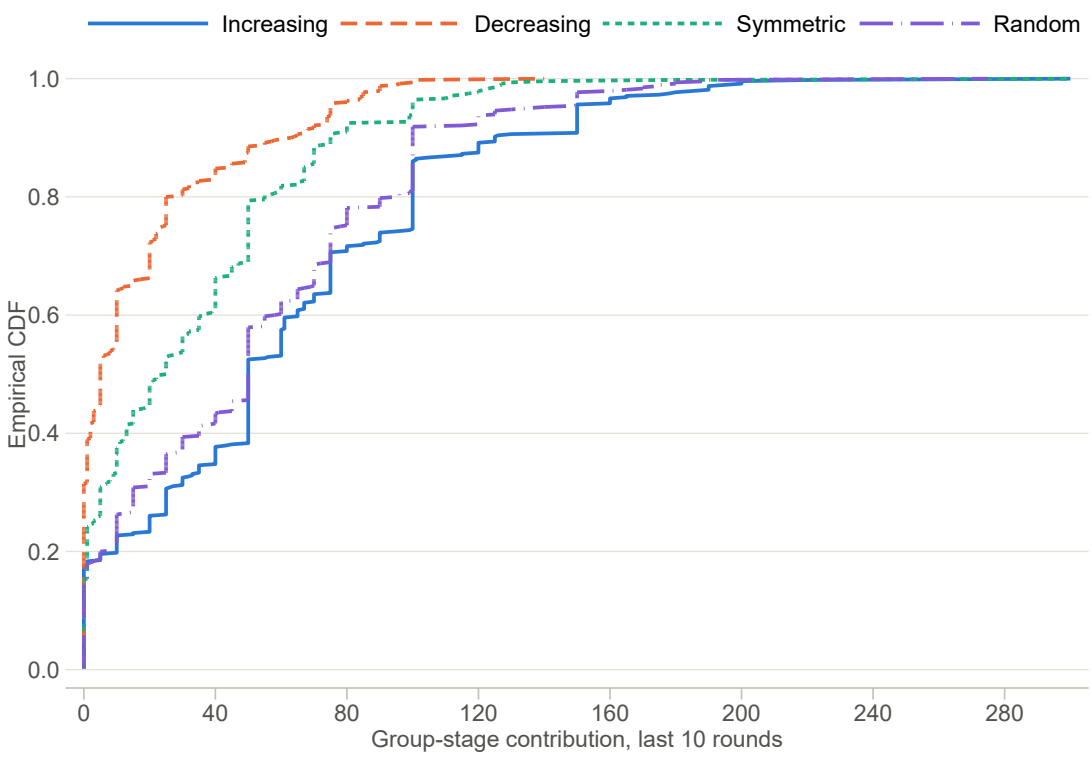


Figure 7: Distribution of Group Stage Contributions, Last 10 Rounds

Notes: Empirical CDF of group stage contributions by treatment. Data from last 10 rounds.

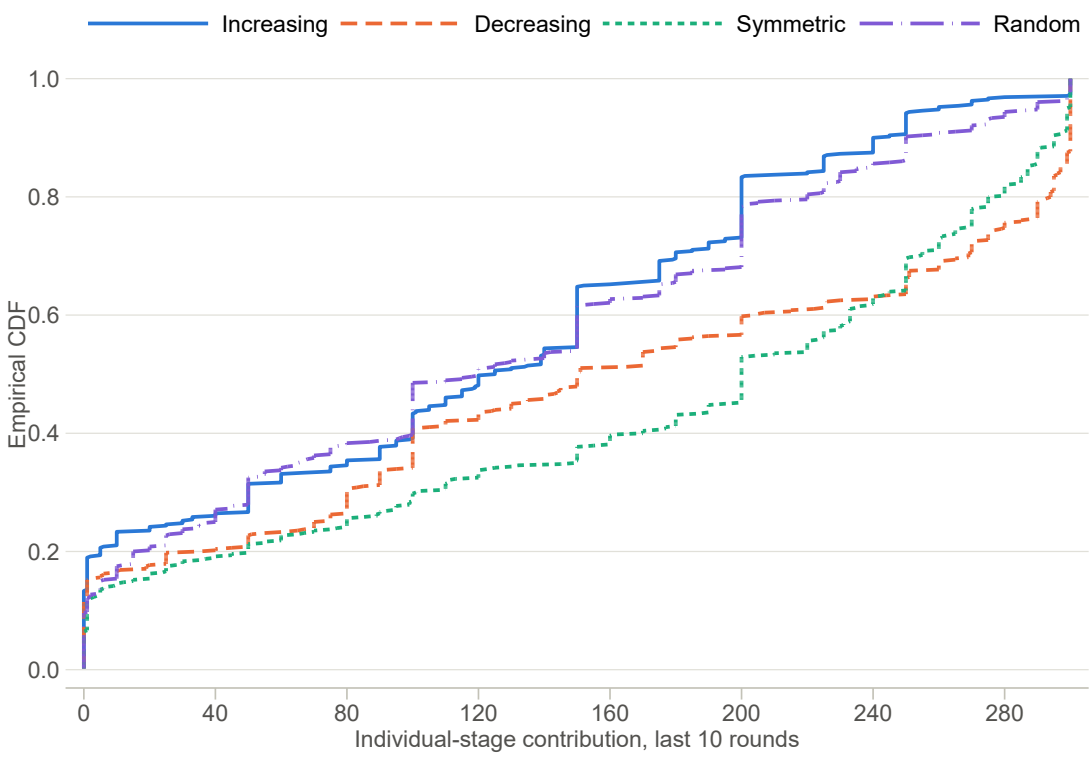


Figure 8: Distribution of Individual Stage Contributions, Last 10 Rounds

Notes: Empirical CDF of individual stage contributions by treatment. Data from last 10 rounds.

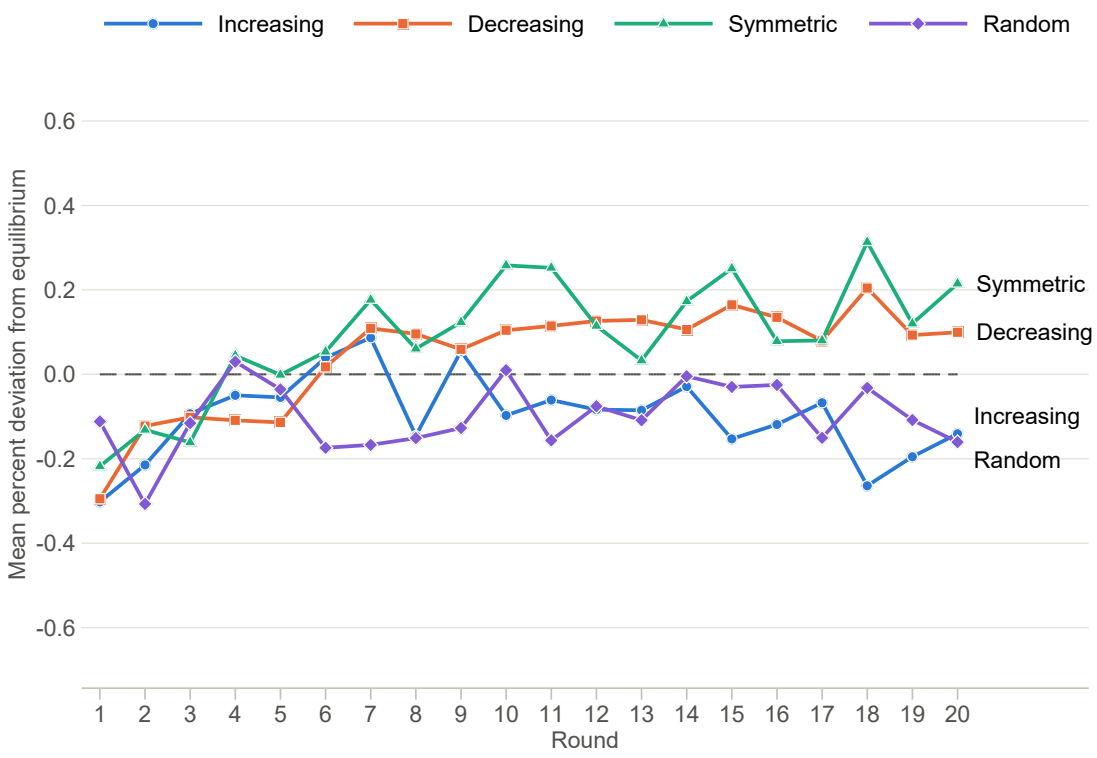


Figure 9: Percent Deviation from Individual Stage Equilibrium (excluding all-in groups)

Notes: Mean percent deviation from individual stage Nash equilibrium in each round by treatment. Dashed line shows theoretical prediction of no deviation. Groups in which at least one subject contributes entire endowment in the group stage are omitted.

Table 6: Group Stage Contributions on Strength for Random Treatment, Last 10 Rounds

	(1)
Strength	18.52**
	[4.940,32.10]
Constant	-3.042
	[-30.70,24.62]
Observations	480
Clusters	6

Notes: Regression of group stage contributions on strength. Data from Random treatment only. Data from last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

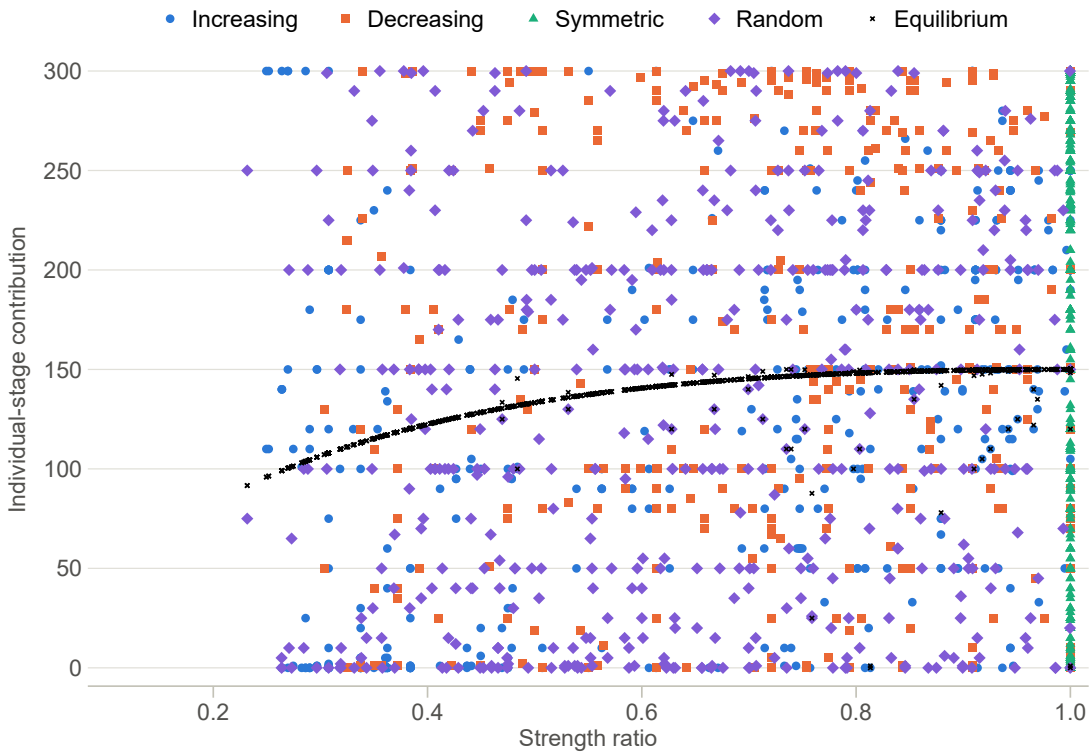


Figure 10: Individual Stage NE by Strength Ratio, Last 10 Rounds

Notes: Raw data on realized strength ratio and corresponding individual stage contribution for each subject by treatment. Black X marks show theoretical individual stage Nash equilibria for given strength ratios. Data from last 10 rounds.

Table 7: Mean Within-Group Total Contributions, Last 10 Rounds

	Mean	Prediction
Increasing	362.754	384.18
	[304.125, 446.582]	
Decreasing	356.413	308.38
	[281.961, 426.815]	
Symmetric	413.850	337.50
	[376.688, 446.219]	
Random	355.800	341.60
	[303.686, 415.649]	
Observations	960	
Clusters	24	

Notes: Mean within-group total contributions over both stages and theoretical predictions by treatment. 95% wild cluster bootstrap confidence intervals in brackets. Data from last 10 rounds.

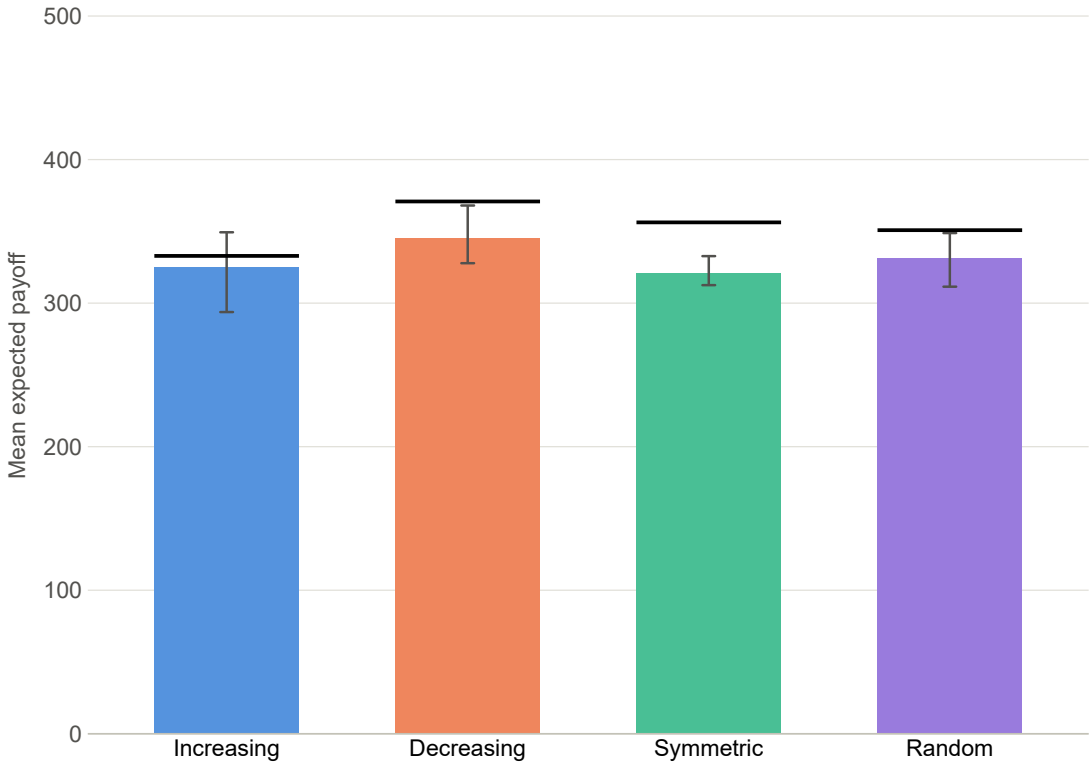


Figure 11: Expected Payoffs, Last 10 Rounds

Notes: Mean expected payoff by treatment given contributions and corresponding probabilities of winning in each stage. Black lines show theoretical predictions. Data from last 10 rounds. Error bars represent 95% wild cluster bootstrap confidence intervals.

Table 8: Mean Expected Payoff, Last 10 Rounds

	Mean	Equilibrium
Increasing	325.036	332.91
	[293.782, 349.353]	
Decreasing	344.885	370.81
	[327.807, 367.983]	
Symmetric	320.917	356.25
	[312.527, 332.746]	
Random	330.990	350.83
	[311.452, 348.806]	
Observations	1920	
Clusters	24	

Notes: Mean expected payoff and theoretical predictions by treatment given contributions and corresponding probabilities of winning in each stage. 95% wild cluster bootstrap confidence intervals in brackets. Data from last 10 rounds.

Table 9: Group Stage Contributions on Lagged Variables

	Increasing	Decreasing	Symmetric	Random
Own Group Stage Contribution	0.671***	0.777***	0.589***	0.462***
	[0.363, 0.811]	[0.529, 0.885]	[0.359, 0.887]	[0.344, 0.536]
Mate Group Stage Contribution	0.023	0.023	0.049	0.048
	[-0.019, 0.099]	[-0.027, 0.145]	[-0.012, 0.103]	[-0.061, 0.107]
Opponent Group Total Contributions	0.053**	0.042***	0.038	0.099***
	[0.002, 0.105]	[0.028, 0.058]	[-0.012, 0.095]	[0.054, 0.126]
Won Group Stage	3.781*	-2.950	-2.438	-0.309
	[-0.722, 8.301]	[-9.488, 2.967]	[-5.909, 1.052]	[-9.700, 10.116]
Won Individual Stage	2.078	4.738**	6.117**	2.337
	[-2.689, 7.004]	[1.250, 10.524]	[2.277, 10.027]	[-3.904, 9.213]
Observations	3648			
Clusters	24			

Notes: Regression of group stage contributions on lagged variables. Round 1 data omitted. 95% wild cluster bootstrap confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Individual Stage Contributions on Lagged Variables

	Increasing	Decreasing	Symmetric	Random
Own Individual Stage Contribution	0.480***	0.513***	0.438**	0.368***
	[0.384, 0.639]	[0.298, 0.788]	[0.300, 0.675]	[0.221, 0.518]
Mate Individual Stage Contribution	0.075**	-0.002	0.082**	0.052
	[0.010, 0.125]	[-0.088, 0.113]	[0.002, 0.132]	[-0.057, 0.138]
Won Group Stage	-6.797	-14.284*	-20.388	-15.759**
	[-21.465, 7.762]	[-29.989, 1.798]	[-45.815, 7.564]	[-30.522, -4.311]
Won Individual Stage	10.716	-9.957	15.701	-3.047
	[-13.452, 37.562]	[-37.182, 24.708]	[-11.179, 38.556]	[-24.748, 12.666]
Current Group Stage Contribution	0.151	-0.036	-0.171	0.097
	[-0.036, 0.549]	[-0.371, 0.418]	[-0.407, 0.182]	[-0.157, 0.427]
Observations	3648			
Clusters	24			

Notes: Regression of individual stage contributions on lagged variables controlling for current period group stage contributions. Round 1 data omitted. 95% wild cluster bootstrap confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

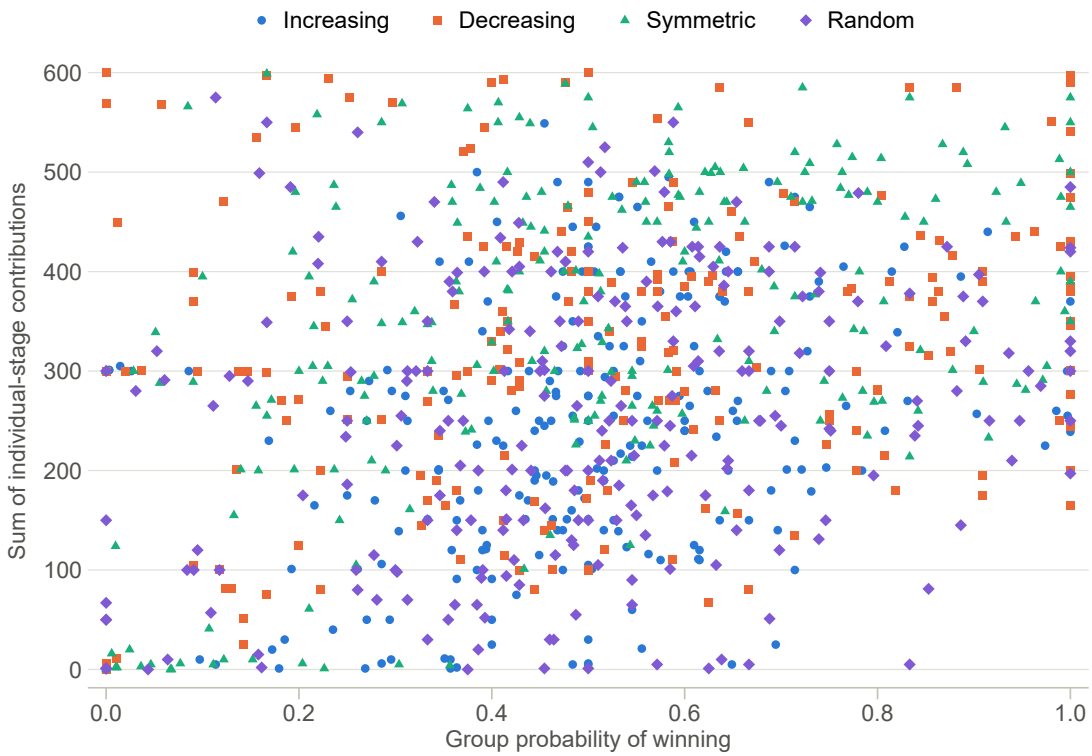


Figure 12: Sum of Ind. Stage Contributions over Group Probability of Winning, Last 10 Rounds

Notes: Raw data on each group's sum of individual stage contributions over their probability of winning the group stage by treatment. Data from last 10 rounds.

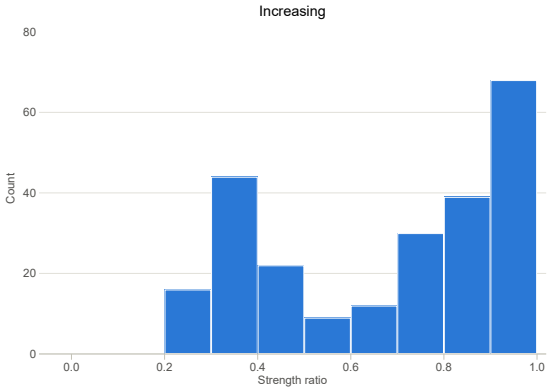


Figure 13: Histogram of Strength Ratios, Increasing, Last 10 Rounds

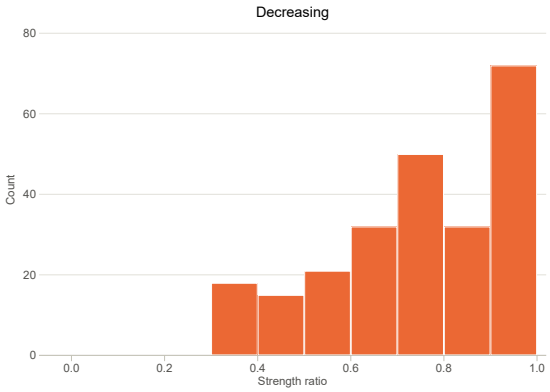


Figure 14: Histogram of Strength Ratios, Decreasing, Last 10 Rounds

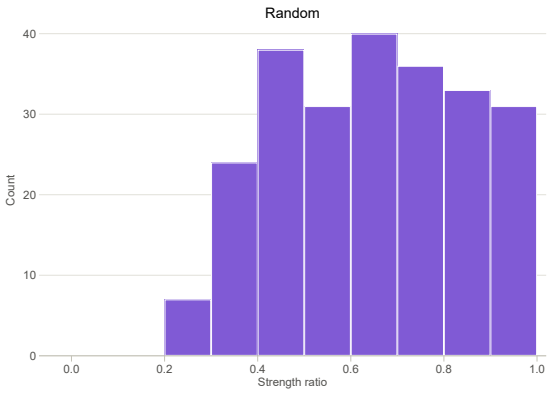


Figure 15: Histogram of Strength Ratios, Random, Last 10 Rounds

Table 11: Individual Stage Contributions on Group Stage Contributions, Last 10 Rounds

	(1)
Increasing	-27.56
	[-71.51,16.39]
Decreasing	42.69*
	[2.305,83.07]
Symmetric	52.52***
	[24.99,80.04]
Group Stage Contribution	0.0836
	[-0.0435,0.211]
Increasing × Group Stage Contribution	0.365
	[-0.275,1.005]
Decreasing × Group Stage Contribution	-0.282
	[-0.777,0.213]
Symmetric × Group Stage Contribution	-0.0812
	[-0.427,0.265]
Constant	121.5***
	[99.97,143.0]
Observations	1920
Clusters	24

Notes: Individual stage contributions on treatment dummies and group stage contributions plus interaction terms. Data from last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 12: Sum of Individual Stage Contributions by Strength Ratio, Last 10 Rounds

	(1)
Increasing	-108.5
	[-275.8,58.70]
Decreasing	148.2
	[-73.40,369.8]
Strength Ratio	117.6
	[-46.89,282.0]
Increasing × Strength Ratio	140.2
	[-88.31,368.7]
Decreasing × Strength Ratio	-120.9
	[-395.6,153.8]
Constant	175.7**
	[65.46,286.0]
Observations	720
Clusters	18

Notes: Sum of individual stage contributions in a group on strength ratio. Random is the omitted category. Data from Symmetric omitted. Data from last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13: Individual Stage Contributions by Treatment and Type, Exact, Last 10 Rounds

Type	Increasing	Decreasing	Symmetric	Random
Equal	133.48	122.90	174.10	.
	[16.08,169.11]	[49.92,177.17]	[151.60,188.34]	
Strong	134.64	182.32	.	138.75
	[109.32,169.79]	[119.49,235.84]		[114.09,165.05]
Weak	105.03	155.18	.	112.97
	[78.40,138.59]	[112.47,205.79]		[88.95,142.33]

Notes: Mean individual stage contributions by treatment and type. Estimates from regressing individual stage contribution on treatment dummy and type dummy. Equal types have exactly equal strengths. Data from last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.

Table 14: Individual Stage Percent Deviation by Treatment and Type, Last 10 Rounds

Type	Increasing	Decreasing	Symmetric	Random
Equal	-0.02	-0.11	0.16	-0.16
	[-0.52,0.19]	[-0.36,0.22]	[0.01,0.26]	[-0.44,0.06]
Strong	0.01	0.29	.	0.03
	[-0.22,0.25]	[-0.18,0.72]		[-0.14,0.23]
Weak	-0.28	0.09	.	-0.18
	[-0.50,0.02]	[-0.22,0.42]		[-0.37,0.02]

Notes: Mean percent deviation from individual stage Nash equilibrium by treatment and type. Estimates from regressing percent deviation on treatment dummy and type dummy. Equal types have exactly equal strengths. Data from last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.

Table 15: Individual Stage Percent Deviation by Treatment and Type, Exact, Last 10 Rounds

Type	Increasing	Decreasing	Symmetric	Random
Equal	-0.11	-0.18	0.16	.
	[-0.89,0.10]	[-0.67,0.18]	[0.01,0.26]	
Strong	0.01	0.29	.	0.01
	[-0.19,0.26]	[-0.17,0.71]		[-0.16,0.21]
Weak	-0.25	0.09	.	-0.18
	[-0.46,0.06]	[-0.21,0.42]		[-0.37,0.02]

Notes: Mean percent deviation from individual stage Nash equilibrium by treatment and type. Estimates from regressing percent deviation on treatment dummy and type dummy. Data from last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.

Table 16: Individual Stage Contributions on Strength Ratio, Type, and Treatment, Last 10 Rounds

	(1)
Increasing	-93.85
	[-220.6,32.89]
Decreasing	30.45
	[-63.72,124.6]
Strong	35.17
	[-17.22,87.55]
Strength Ratio	66.04
	[-34.90,167.0]
Increasing × Strong	63.42
	[-26.96,153.8]
Decreasing × Strong	-3.616
	[-276.9,269.7]
Increasing × Strength Ratio	132.6
	[-26.24,291.4]
Decreasing × Strength Ratio	12.56
	[-113.0,138.1]
Strong × Strength Ratio	-14.53
	[-76.71,47.65]
Increasing × Strong × Strength Ratio	-92.01*
	[-175.5,-8.495]
Decreasing × Strong × Strength Ratio	8.168
	[-324.7,341.1]
Constant	70.29
	[-1.094,141.7]
Observations	1314
Clusters	18

Notes: Individual stage contributions on strength ratio, type and treatment dummies. Random and Weak are the omitted categories. Only data on groups with asymmetric strengths included. Data from Symmetric omitted. Data from last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 17: Slope Coefficients on Strength Ratio by Type, Last 10 Rounds

	Increasing	Decreasing	Random
Weak	198.608***	78.604*	66.043
	[42.470, 313.848]	[-0.715, 151.050]	[-67.126, 185.091]
Strong	92.069**	72.244	51.514
	[2.399, 161.794]	[-230.165, 355.439]	[-47.097, 146.413]
Observations		1314	
Clusters		18	

Notes: Slope coefficients on strength ratio from regression of individual stage contributions on strength ratio, type, and treatment. Data from last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 18: Strong Player Share of Individual Stage Contributions on Strength Ratio, Last 10 Rounds

	(1)
Increasing	0.327*
	[0.0774,0.577]
Decreasing	-0.259
	[-0.849,0.332]
Strength Ratio	-0.145
	[-0.291,0.00176]
Increasing × Strength Ratio	-0.436**
	[-0.690,-0.182]
Decreasing × Strength Ratio	0.242
	[-0.475,0.959]
Constant	0.694***
	[0.577,0.810]
Observations	657
Clusters	18

Notes: Stronger player's share of total individual stage contributions in a group on strength ratio and treatment dummies plus interaction terms. Random is the omitted category. Only data on groups with asymmetric strengths included. Data from Symmetric omitted. Data from last 10 rounds. 95% wild cluster bootstrap confidence intervals in brackets.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

C Experiment Instructions (Increasing Treatment)

Introduction

Thank you for participating in today's experiment. You will be participating in a type of market in which your earnings will depend on your own decisions and the decisions of other participants. Your total earnings, including your $7 show-up fee, will be paid to you privately at the end of the experiment. Please remain quiet and do not communicate with other participants.

We would not want anyone's earnings to suffer because you did not understand the rules of how these markets operate. If you have a question or a problem during the experiment, we encourage you to raise your hand and one of the experimenters will do their best to try and answer it for you. We cannot tell you what a good or bad decision is for you, but we will gladly help explain the rules.

In the experiment, your payoffs will be denominated in points. We will explain in a moment how points are converted into U.S. dollars. All amounts in this experiment are expressed in points. The exchange rate is 1 point = $0.01. This part of the experiment consists of a sequence of 20 decision rounds.

Pairing and matching

At the beginning of the experiment, you will be randomly placed into a pool of 8 participants. You will remain in this pool for the duration of the experiment, and you will only interact with the other members of your pool.

Within each round, you will be paired in a group with one other participant. Your group will then be matched with another group of two different participants. Between rounds, you'll be randomly re-paired with another group member and randomly rematched with another group.

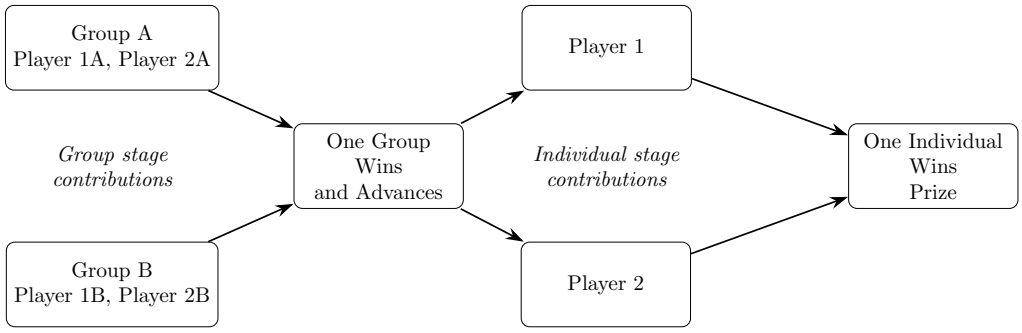
Endowment

At the beginning of each round, you will be given an endowment of 300 points. You may use any number of these points to make decisions in a given round.

Round Overview

Each round consists of two stages: the group stage and the individual stage. In the group stage, two groups will compete to determine which group advances to the individual stage. In the individual stage, members of the winning group will compete with each other for a prize. Your task will be to contribute points from your endowment to the competition in each stage. Ultimately, only one out of the four players that enter a match in each round will win the prize. This player must belong to the winning group and win the individual stage.

The following flow chart summarizes how each round will proceed:



Payoffs

The payoff from each round will depend upon the outcomes in both stages in the following way:

$$\text{Round payoff} = \underbrace{300}_{\text{Endowment}} - \text{group stage contribution} - \underbrace{\text{individual stage contribution}}_{\text{only if group wins}} + \underbrace{600}_{\substack{\text{Prize}\\ \text{only if group wins}\\ \text{\& individual wins}}}$$

In each round, you earn your endowment of 300 points minus the points you contribute to the group stage. If your group wins, you also pay the points you contribute to the individual stage. Finally, if you are the one individual who wins the prize, you receive 600 points.

Group Stage

In the group stage, two groups of two players will compete to determine which group advances to the individual stage. You will decide how many points you wish to contribute to your group. You can contribute any integer number between 0 and 300 (0 and 300 inclusive). Any points which you don't contribute in this stage will remain with you for the next stage. Your group's total contributions will be the sum of the contributions of you and your group mate.

You can't earn any additional points in the group stage. You compete only for the right to advance to the individual stage where you will have the opportunity to compete for a prize.

What is the likelihood that your group wins?

Only one of the two groups can win. Your group's likelihood of winning is computed as the share of your group's contributions in the total contributions from both groups. The **probability that your group wins** is given by:

$$\frac{\text{your group's contributions}}{\text{your group's contributions+other group's contributions}}$$

If all players contribute 0 points, then the probability that your group wins is $\frac{1}{2} = 50.00\%$.

Once every player has submitted their contributions, the computer program will use the probabilities of winning to determine which group advances to the individual stage.

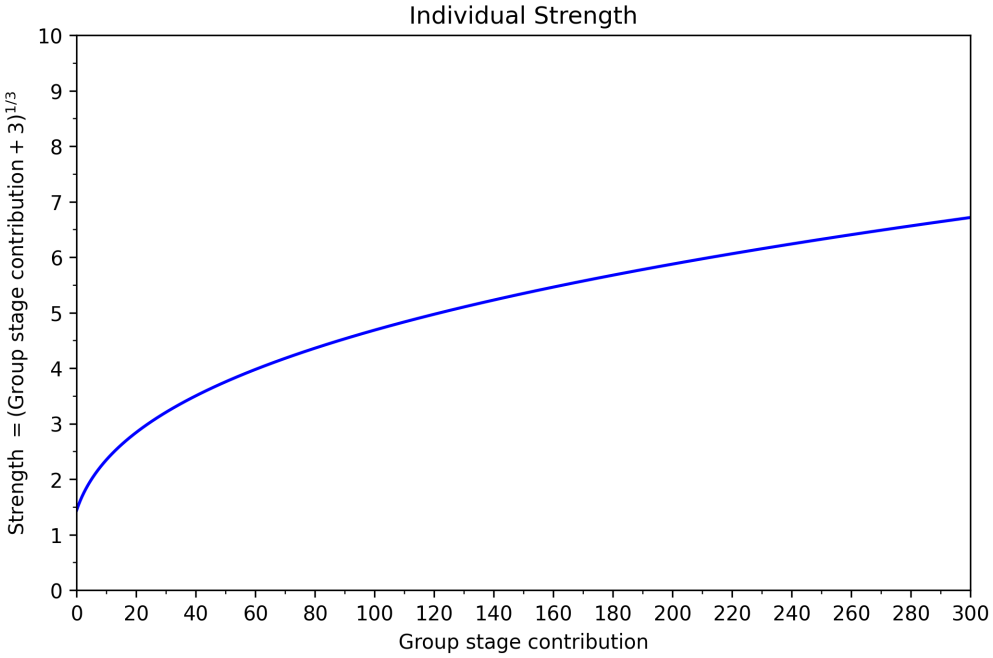
Individual Stage

In the individual stage, you will compete with your group mate for a prize of 600 points. At this point, you won't know whether your group won or lost in the group stage. You will make your decision in the individual stage as if your group won.

You will choose how many points you wish to contribute to the individual competition. You can choose any integer number less than or equal to your **remaining endowment**. Your **remaining endowment** is the share of your endowment that you didn't contribute in the group stage (300 − group stage contribution). The more points you contribute in the group stage, the fewer points you have to contribute in the individual stage.

Individual strength and multiplied contribution

Your contribution in this stage will be multiplied by your individual **strength** which depends on your group stage contribution. The following figure illustrates how group stage contribution determines strength:



Notice, the *more* points you contribute in the group stage, the *higher* your strength will be in the individual stage. Your group mate's individual stage contributions will be multiplied by his/her strength which depend on his/her group stage contributions in the same way.

Your multiplied contribution is the product of your individual stage contribution and your strength:

$$\text{Multiplied contribution} = \text{your individual stage contribution} \times \text{your strength}$$

What is the likelihood that you win the individual stage? Only one of you or your group mate can win the individual stage. The likelihood that you win is the share of your multiplied contribution in the total multiplied contributions of both you and your group mate. The **probability that you win** is given by:

$$\frac{\text{Your multiplied contribution}}{\text{Your multiplied contribution} + \text{your group mate's multiplied contribution}}$$

If you and your group mate both invest zero, then the probability that you win is:

$$\frac{\text{Your strength}}{\text{Your strength + your group mate's strength}}$$

Once every player has submitted their contributions, the computer program will use the probabilities of winning to determine which individual wins the prize.

How are your earnings determined?

You will participate in a series of 20 decision rounds. Before the experiment began, **four** of these rounds were chosen randomly (with all rounds being equally likely to be chosen) to determine your earnings. At the end of the experiment, you will be informed about which four rounds were chosen and your payoff from each of those four rounds. Then your earnings from the experiment will be the sum of your payoffs from the four randomly selected rounds.